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One-minute frequency disturbances from unintended synchronization in cyber–physical power systems

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Supplementary Information for One-minute frequency disturbances from unintended synchronization in cyber–physical power systems

Carsten Hartmann [1,2] Edoardo De Din [1] Daniele Carta [1] Florian Middelkoop,[3] Arndt Neubauer,[4] Johannes Kruse [1,3] Ulrich Oberhofer [5] Richard Jumar [5] Benjamin Schäfer [5] Thiemo Pesch [1] Andrea Benigni [1,2,6] and Dirk Witthaut [1,3,6]

[1] *Institute of Climate and Energy Systems: Energy Systems Engineering (ICE-1), Forschungszentrum Jülich, 52428 Jülich, Germany*

[2] *RWTH Aachen University, 52062 Aachen, Germany*

[3] *Institute for Theoretical Physics, University of Cologne, 50937 Köln, Germany*

[4] *TenneT TSO GmbH, Bernecker Str. 70, 95448 Bayreuth, Germany*

[5] *Institute for Automation and Applied Informatics, Karlsruhe Institute of Technology, 76344 Eggenstein-Leopoldshafen, Germany*

[6] *JARA-ENERGY, 52074 Aachen, Germany*

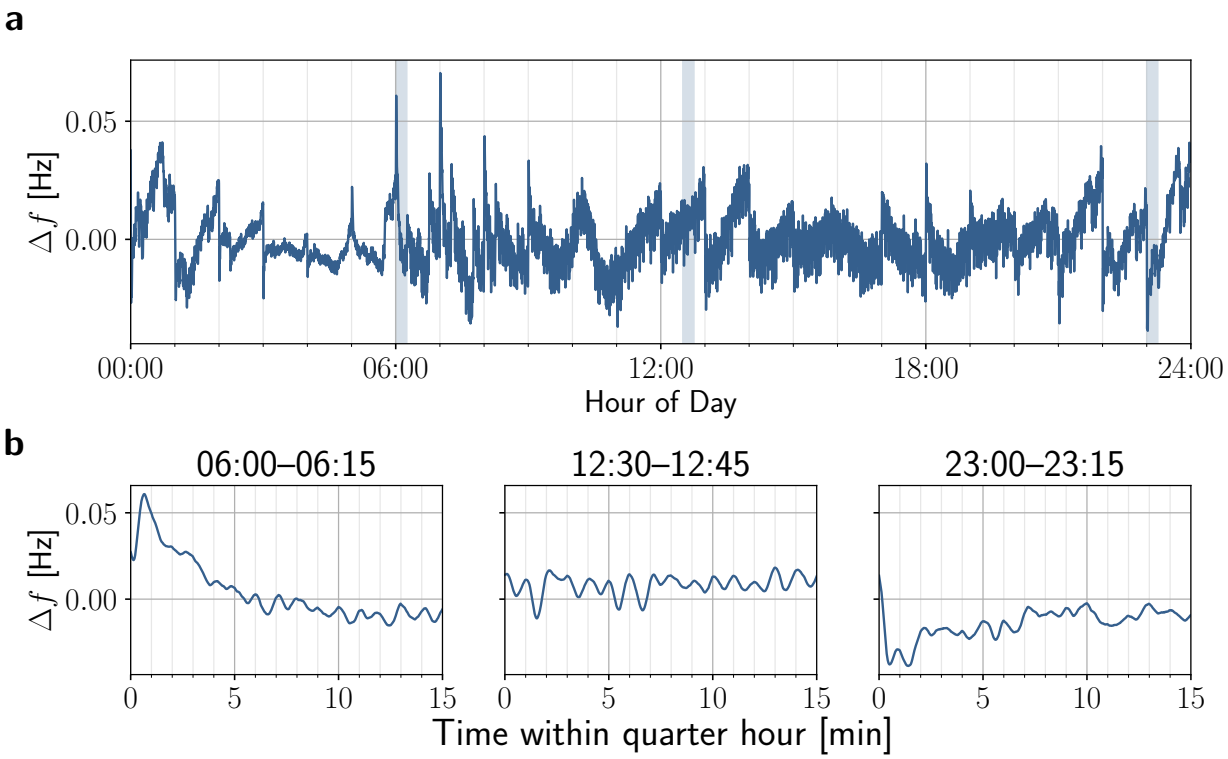
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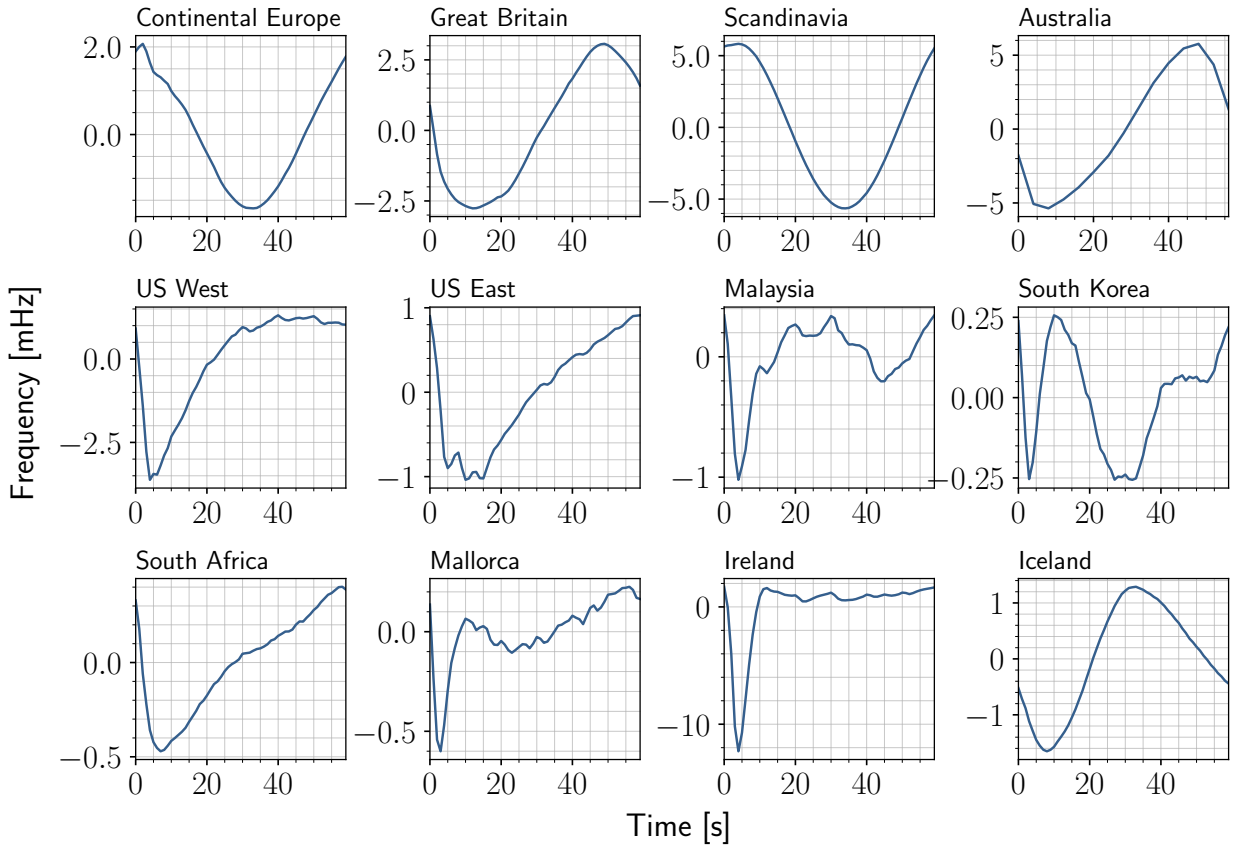
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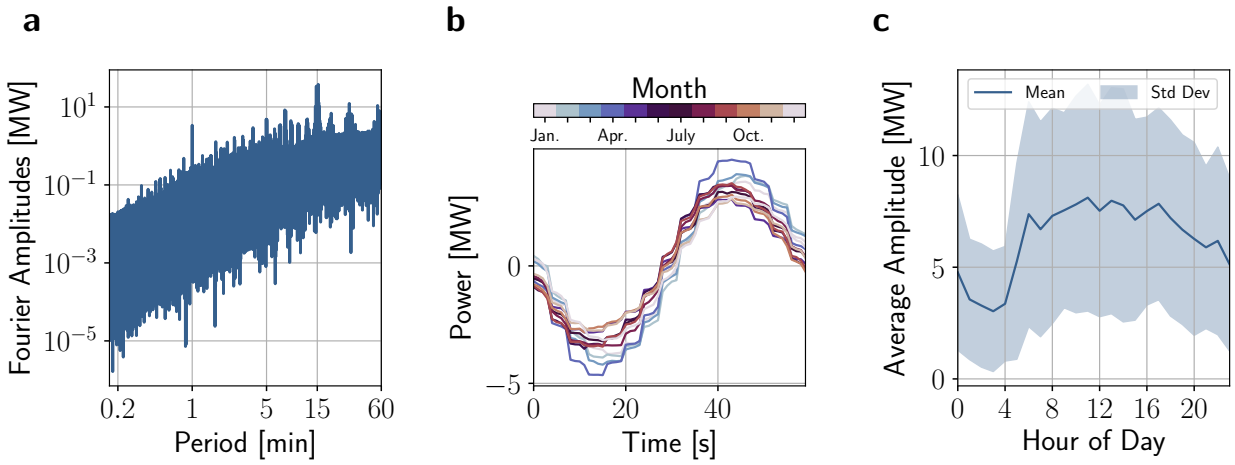
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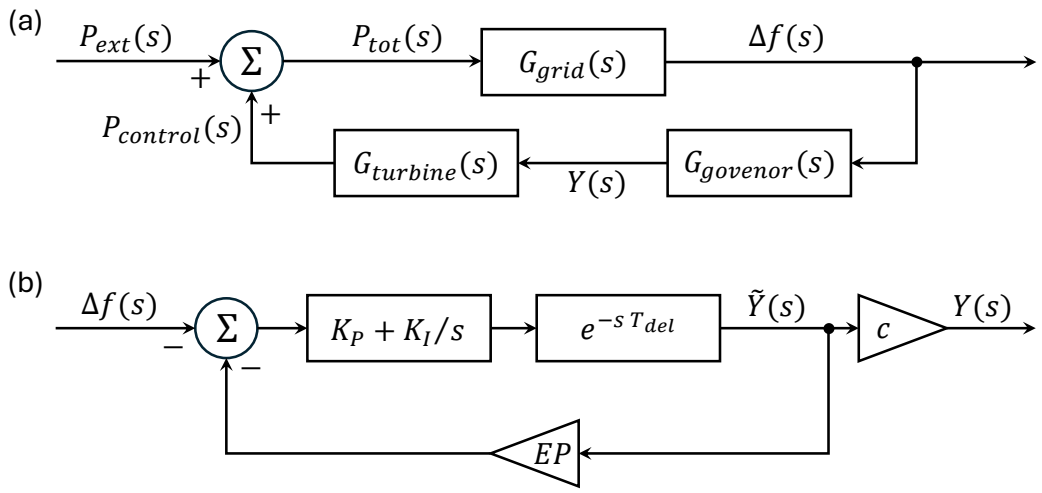
Supplementary Figure 1. Daily profile of the frequency deviation in the Scandinavian grid in 2019. **a**, Averaged frequency deviation (daily profile), obtained by averaging over all timestamps with the same local time throughout the year. Discrete Frequency Deviations (DFDs) are clearly visible at the beginning of each hour. **b**, Zoom into three 15-minute intervals highlighted in panel. The left panel shows a DFD caused by a positive generation ramp, the middle panel shows the frequency behaviour in the middle of an hour, and the right panel shows a DFD caused by a negative generation ramp. In all cases, a stable phase-coherent oscillatory pattern with a period of 1 min is clearly visible.



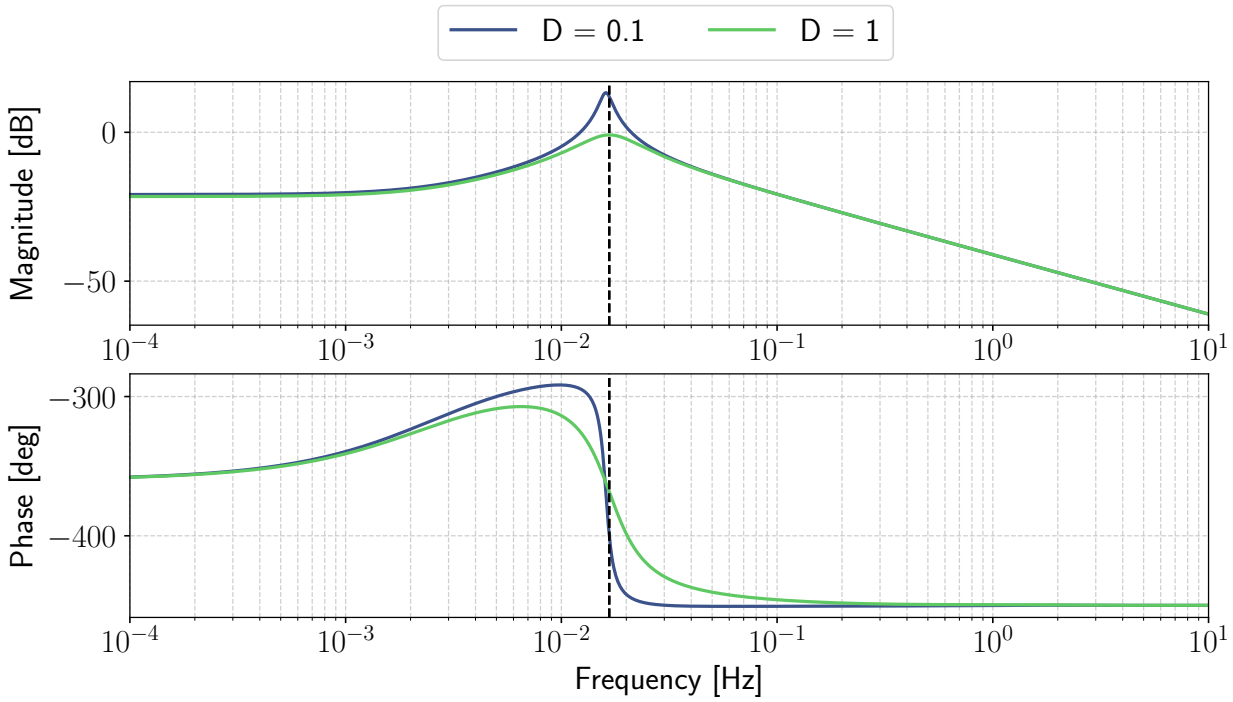
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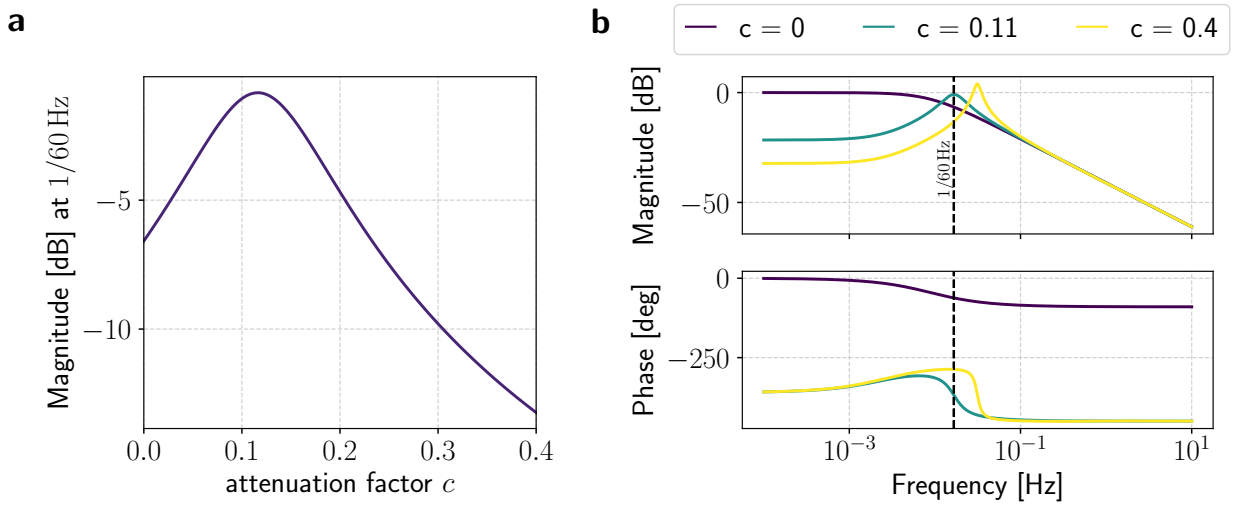
Supplementary Figure 3. The 1 min pattern in the secondary control setpoint in Germany. We analyse the time series of the automatic frequency restoration reserve (aFRR) setpoint in the German grid for the year 2025. The data are publicly available at 1 s resolution [1]. **a**, The Fourier spectrum of the aFRR setpoint exhibits a narrow peak at a period of 1 min. **b**, The monthly waveforms are approximately sinusoidal, with negative values during the first half and positive values during the second half of the minute. Furthermore, a 4 s pattern is visible, possibly related to the aFRR setpoint calculation process. **c**, Daily profile of the signal amplitude, which closely follows the corresponding pattern in the frequency signal (cf. Main Text). The solid line shows the mean hourly amplitude, and the shaded area indicates ±1 standard deviation.



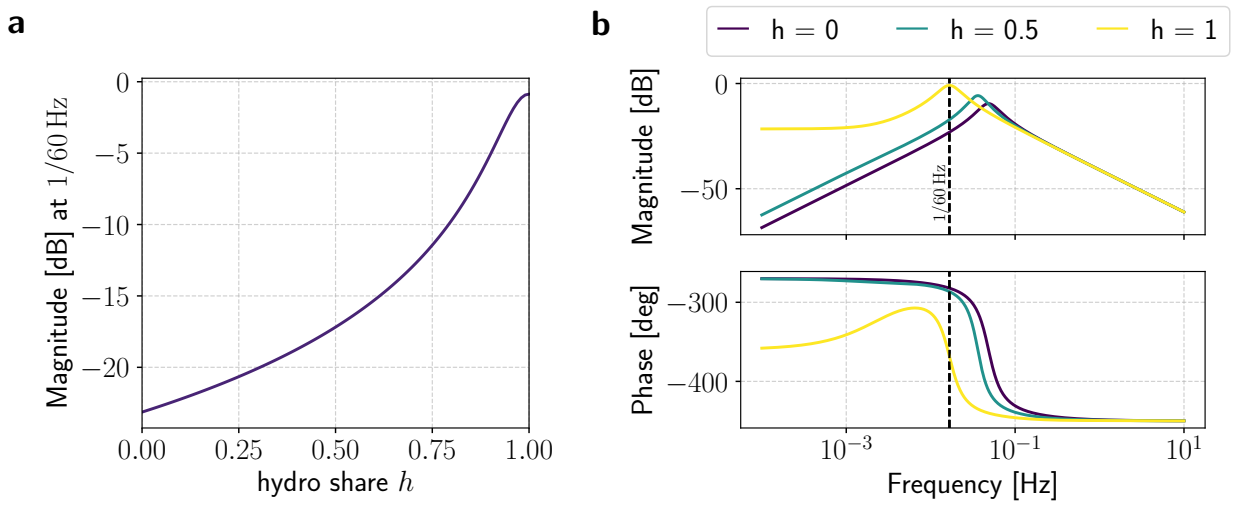
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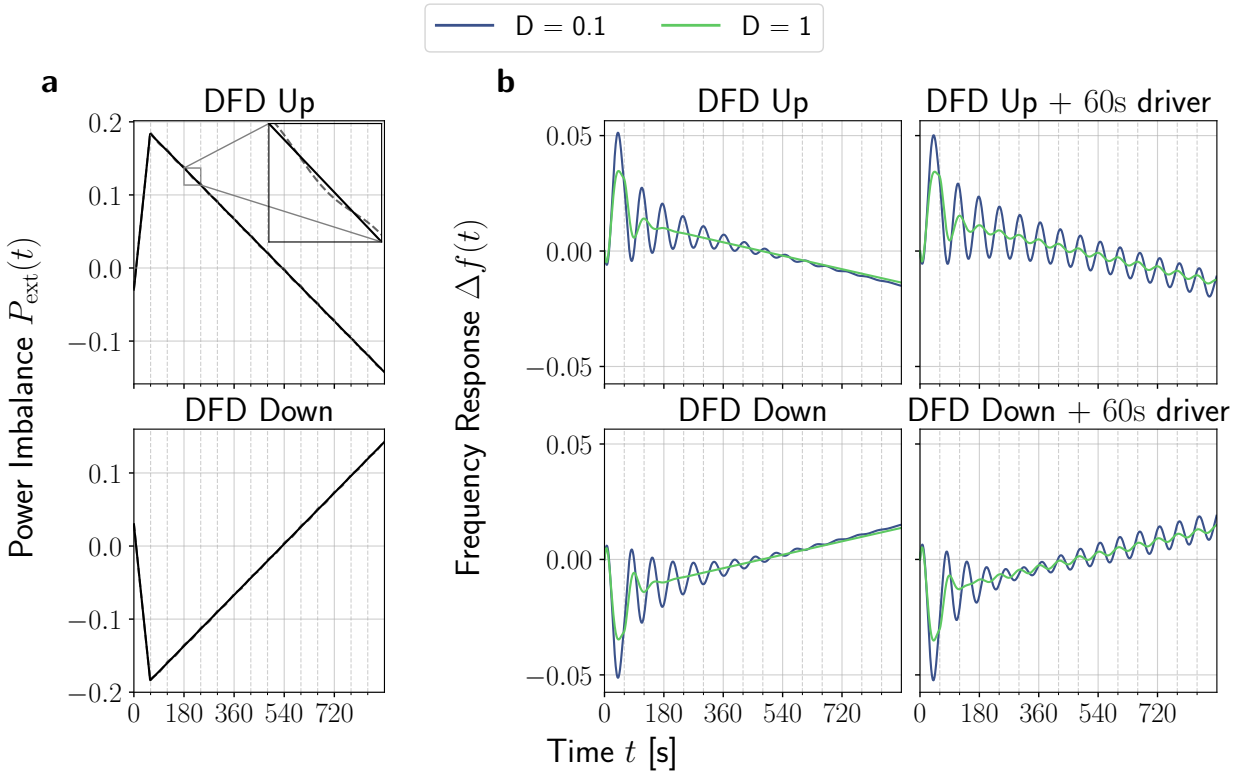
Supplementary Figure 5. Bode plot of the closed loop transfer function (2). A resonance at a frequency of approximately 1/60 Hz emerges for a wide range of grid damping values D due to a turbine–governor resonance in hydro power plants providing load-frequency control.



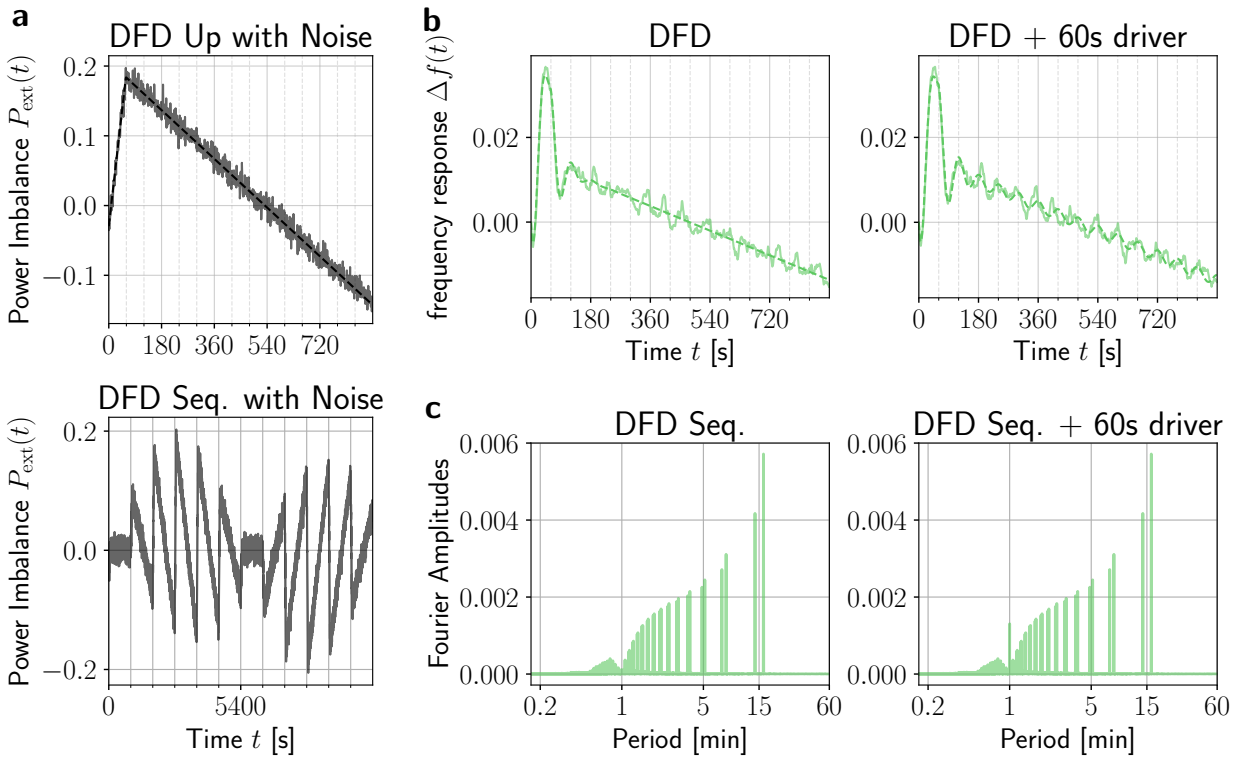
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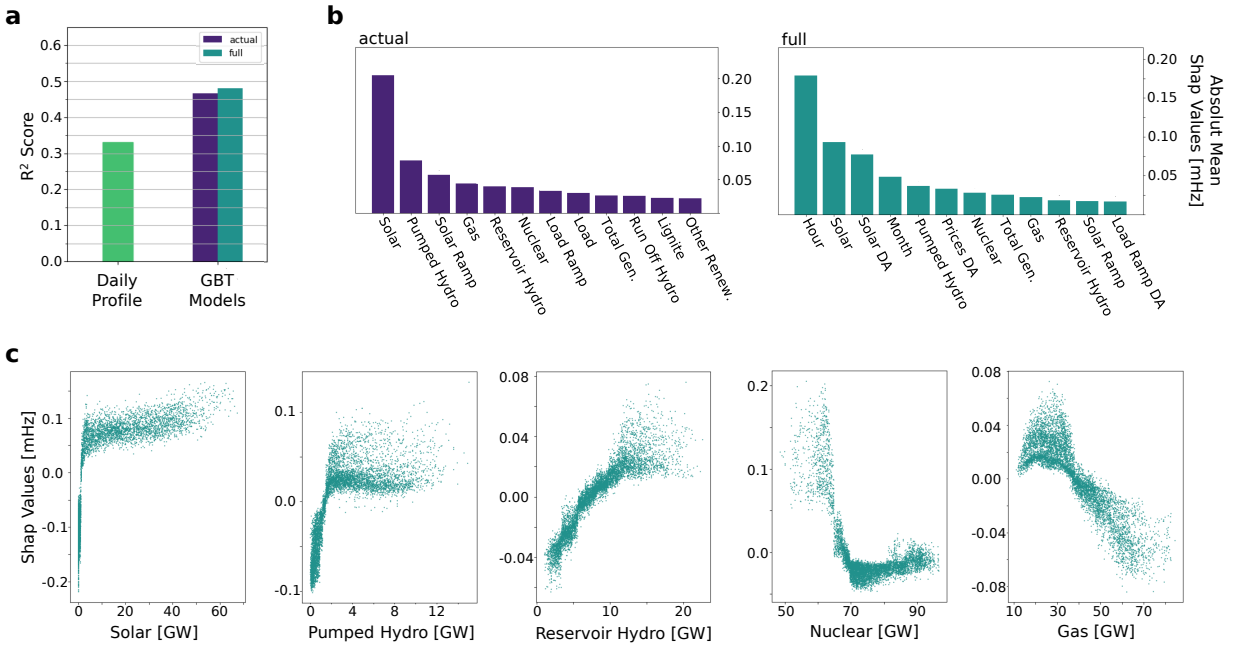
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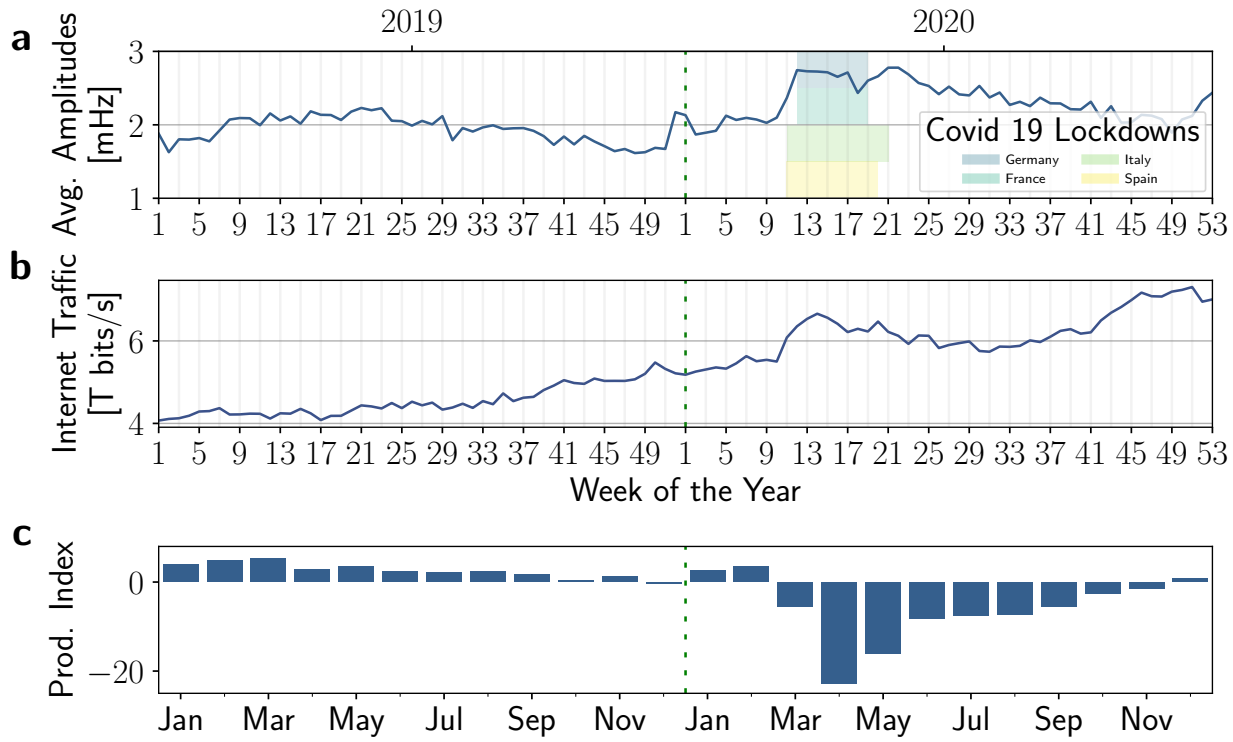
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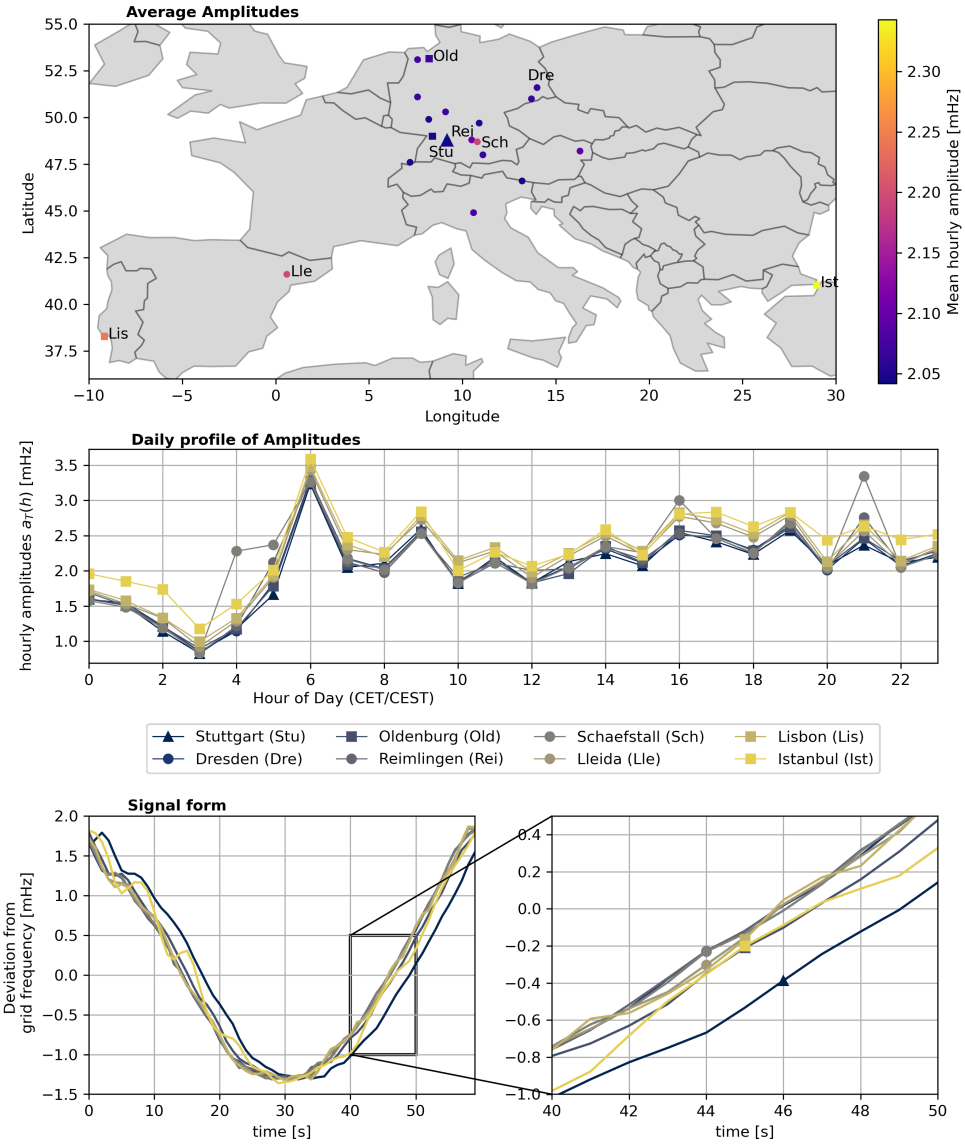
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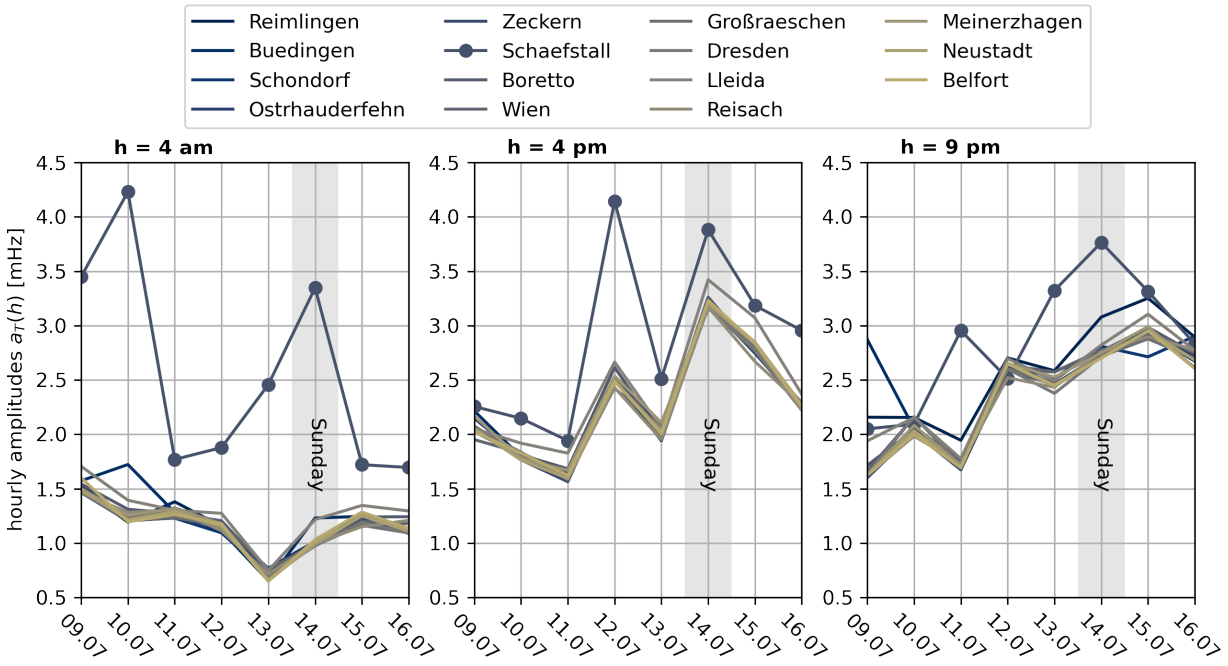
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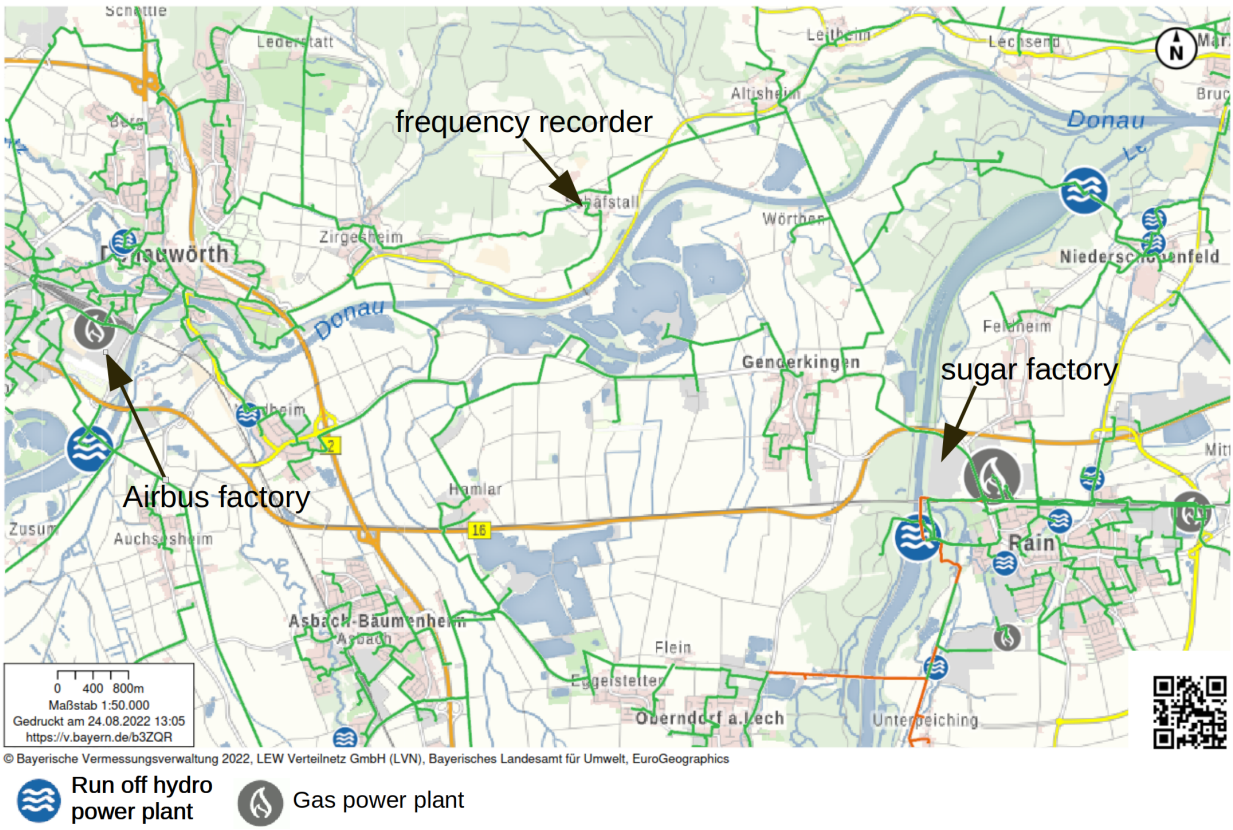
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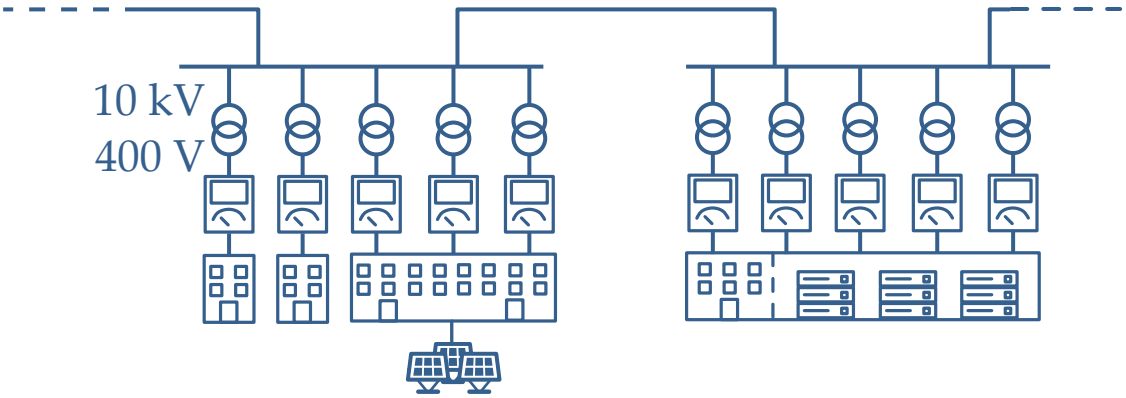
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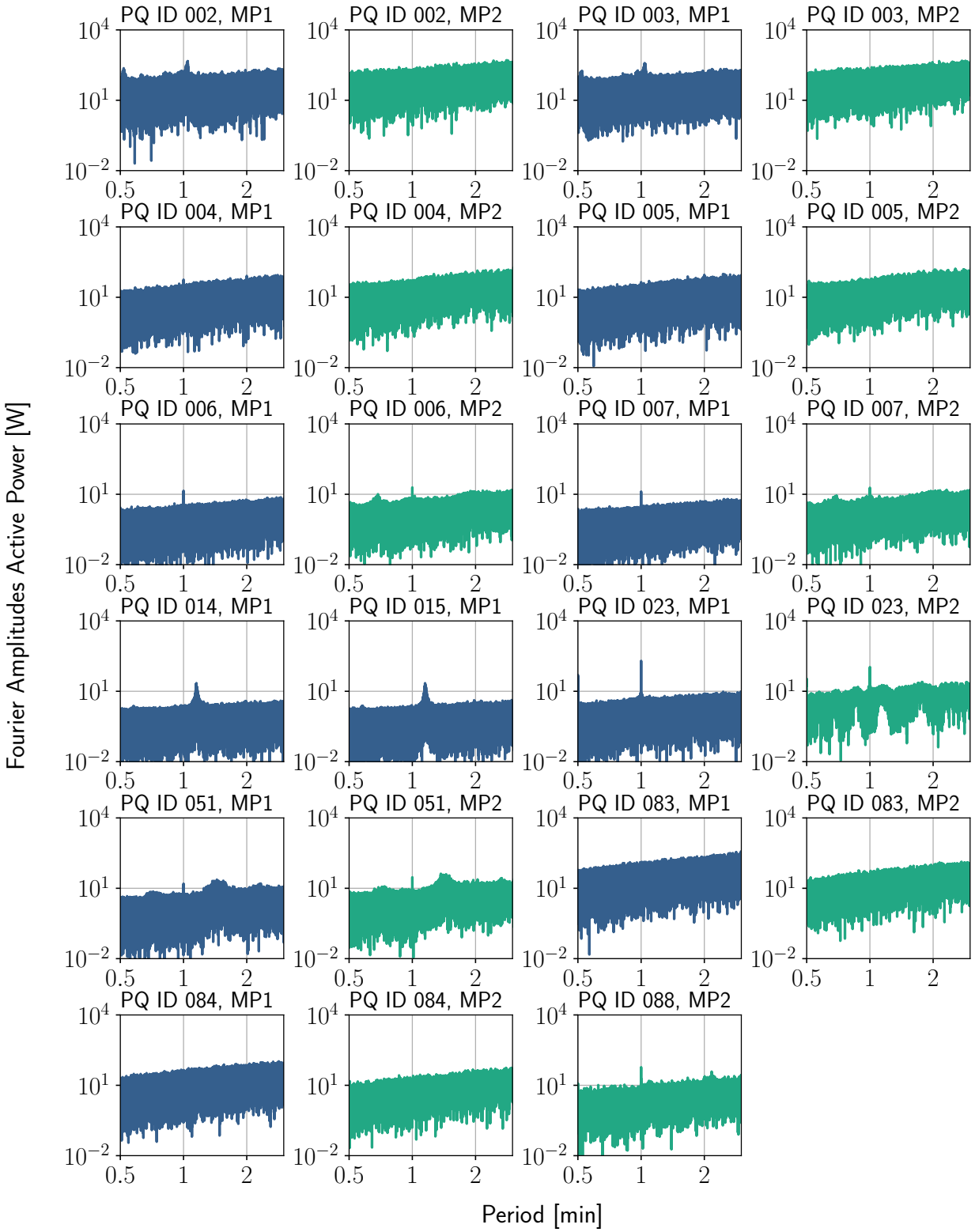
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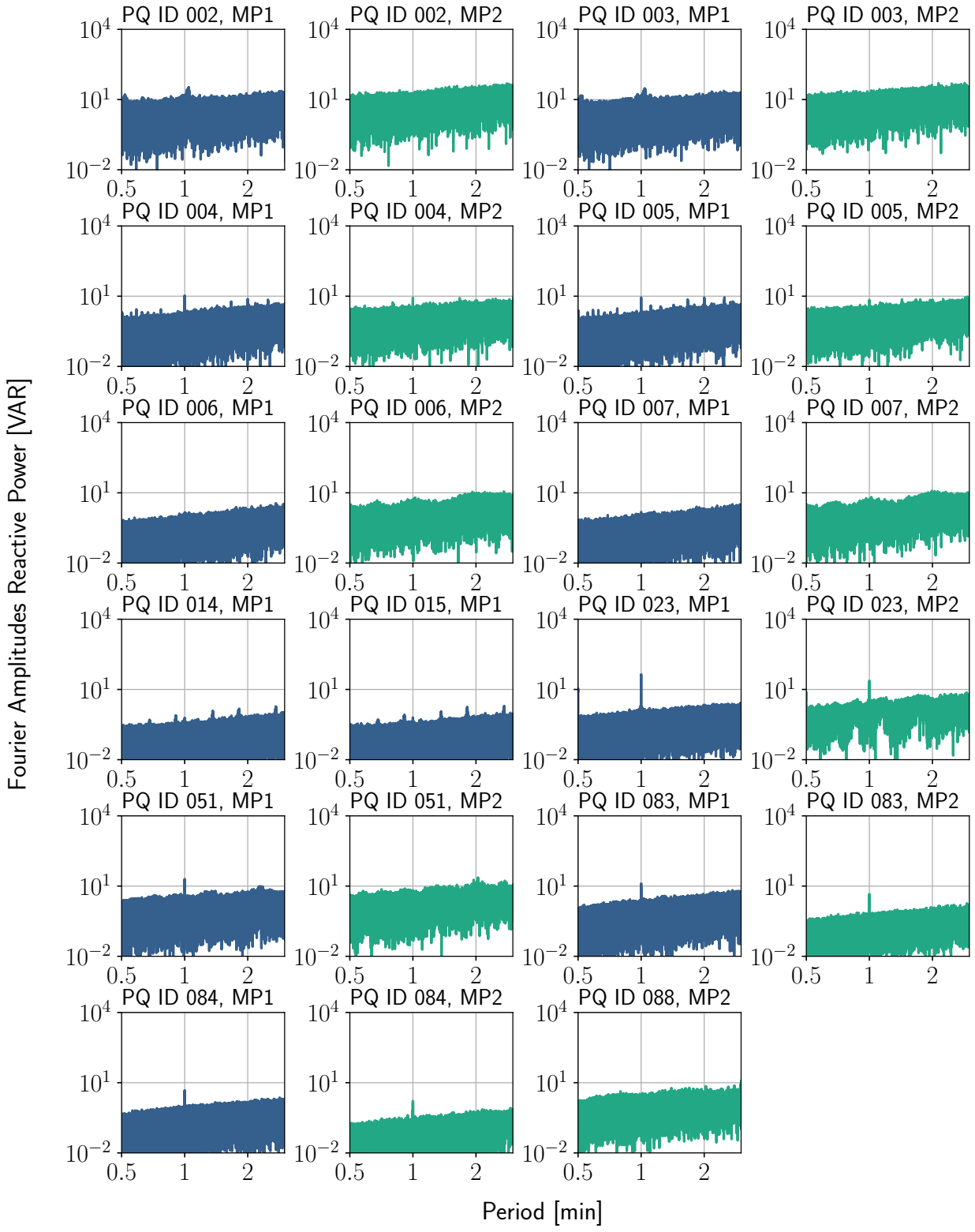
Supplementary Figure 14. Map of points of Interest around Schaefstall, reproduced from Energie Atlas Bayern [7] with permission. There are several factories, gas power plants and run-of-river hydro power plants that could be potential local drivers or amplifiers of the 1 min pattern.



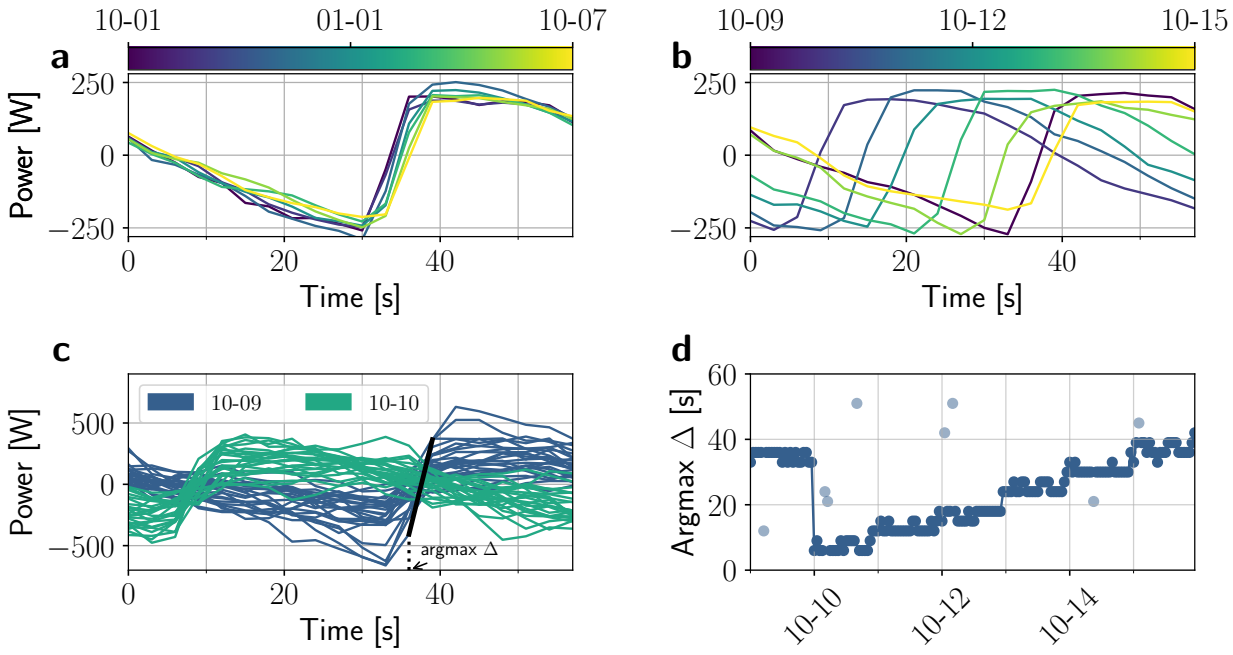
Supplementary Figure 15. Schematic representation of the measurement infrastructure of the FZJ Campus.



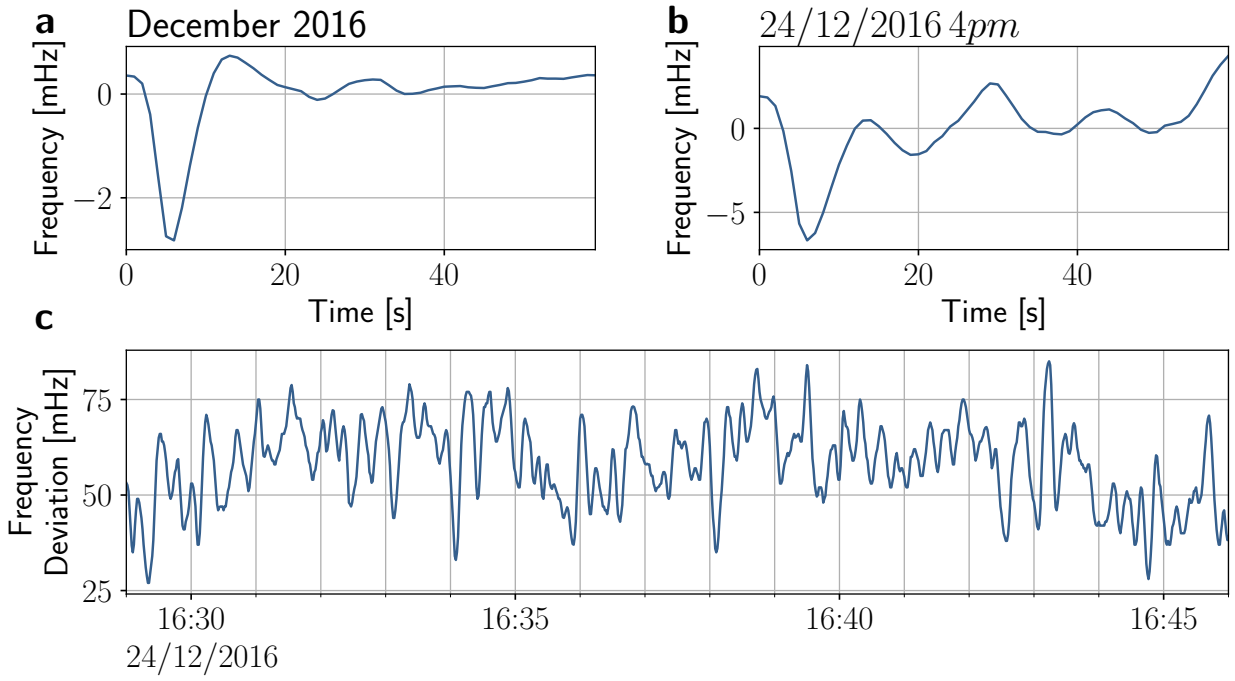
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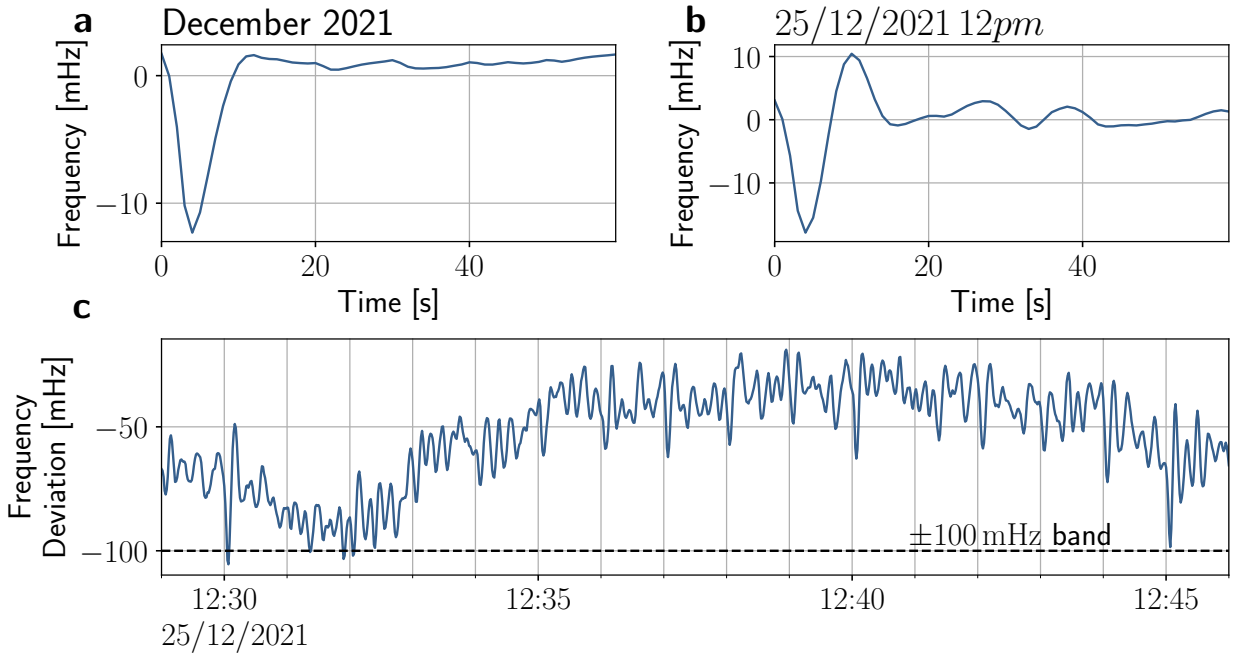
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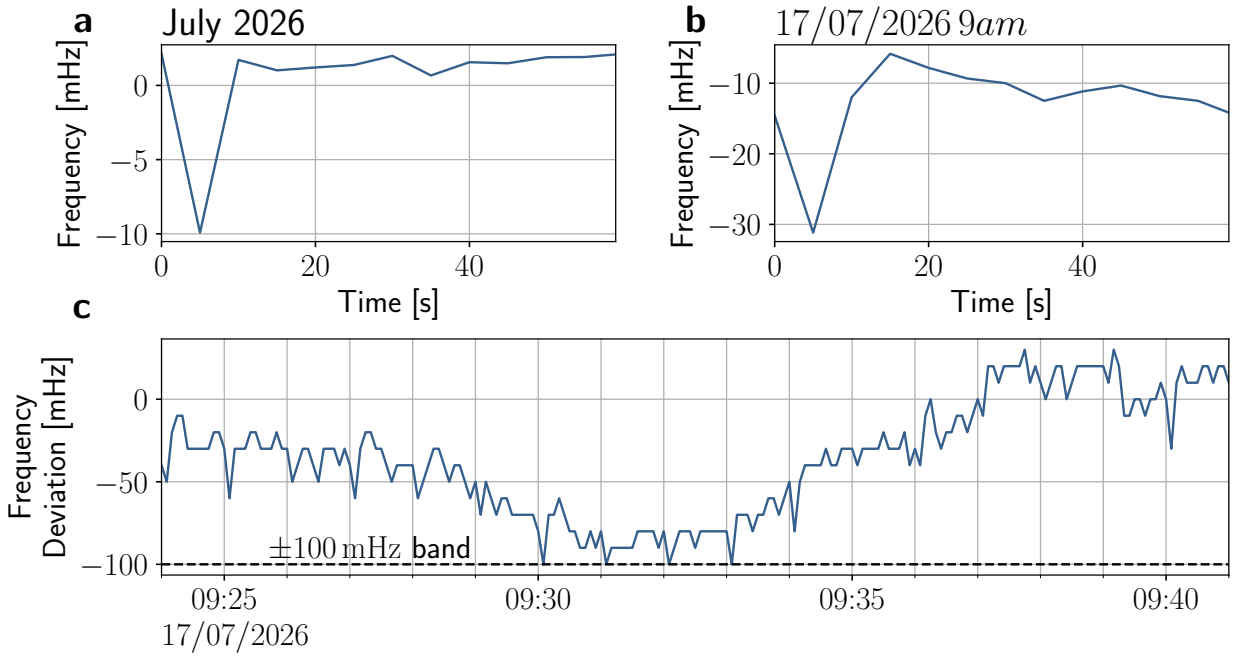
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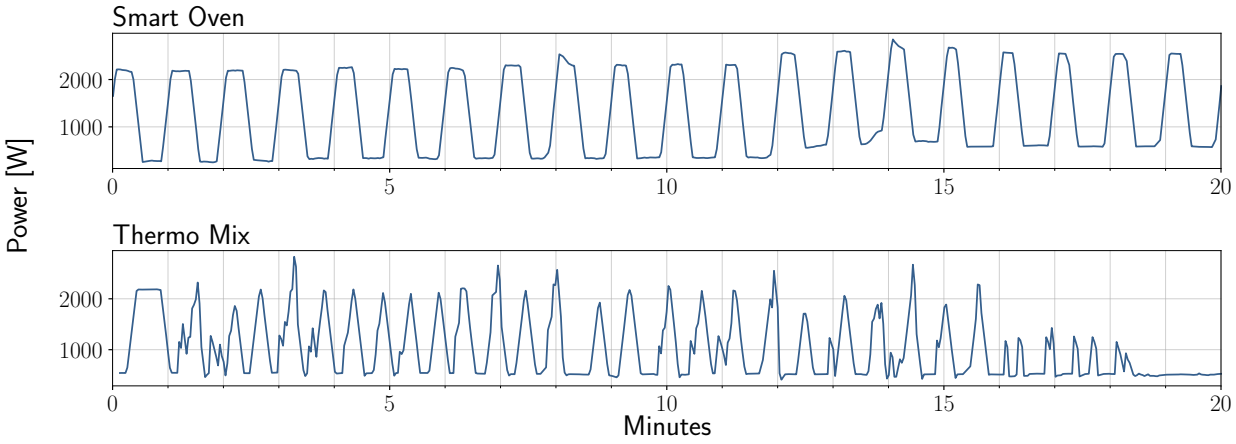
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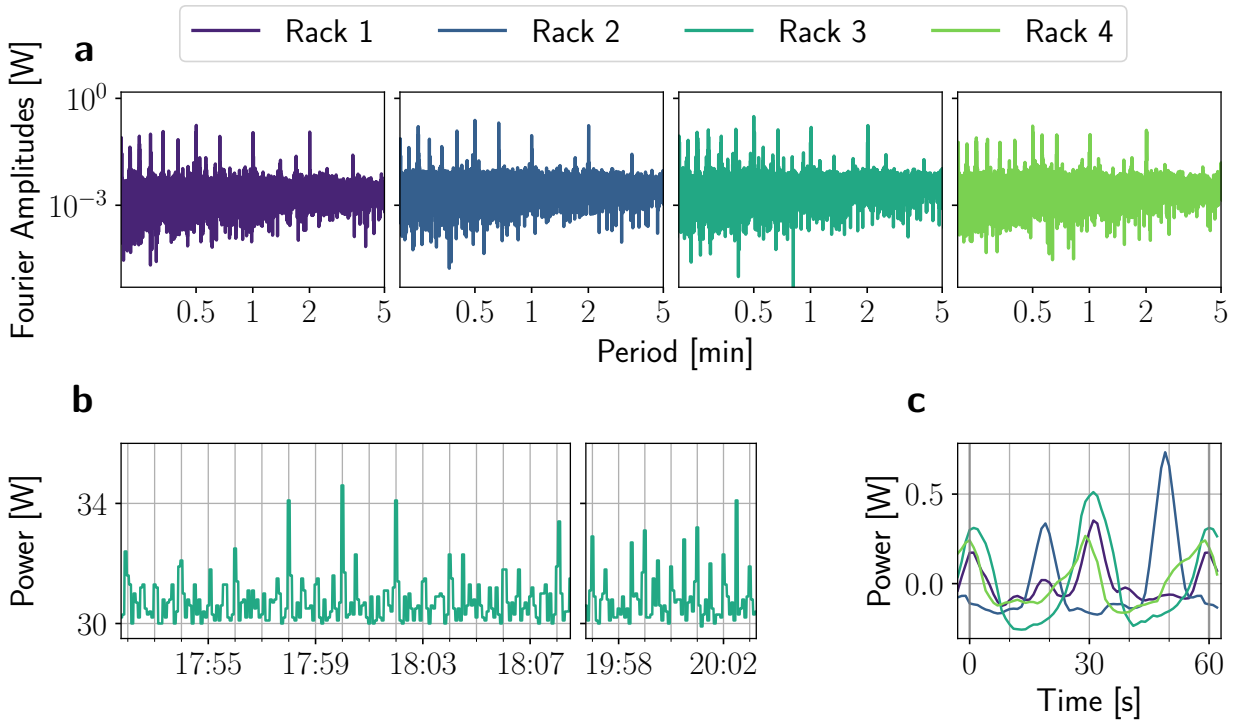
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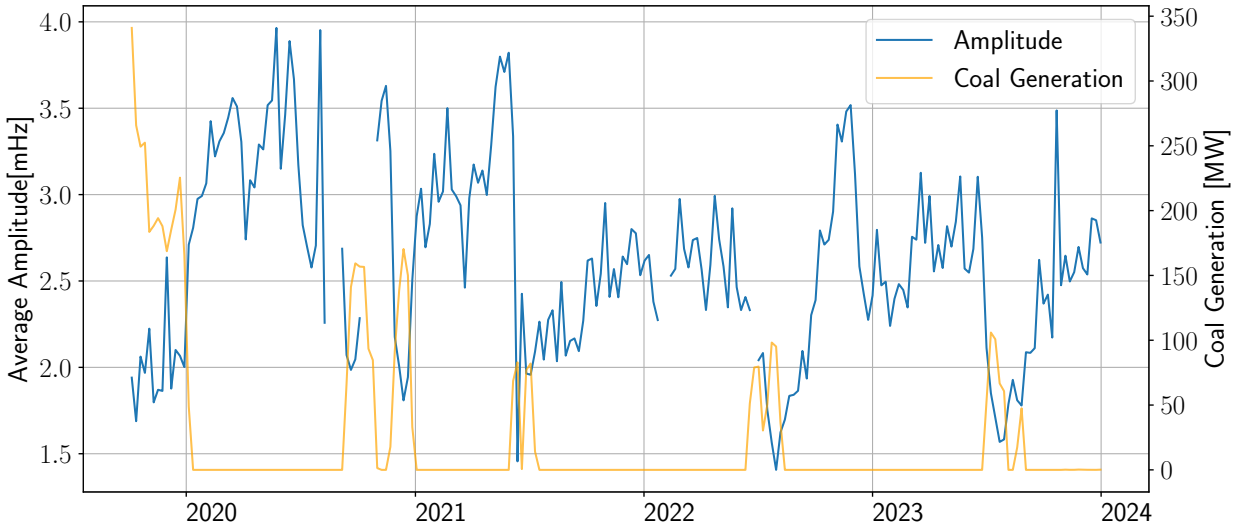
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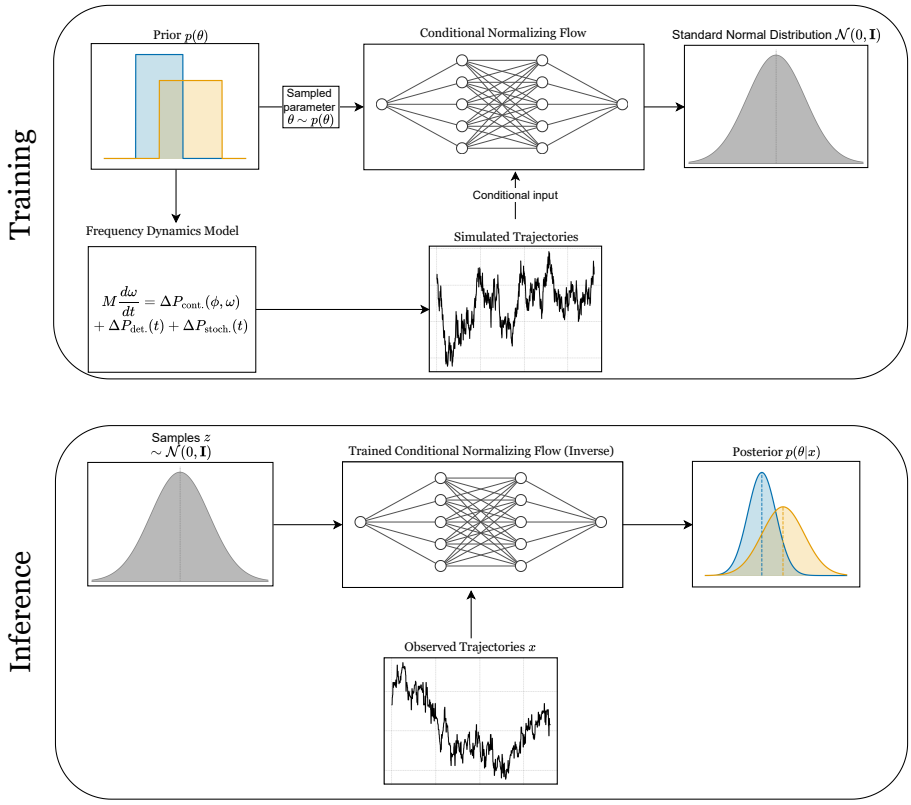
Supplementary Figure 22. Time series of the active power consumption for two smart devices: a smart baking oven (top) and a Thermomix (bottom).



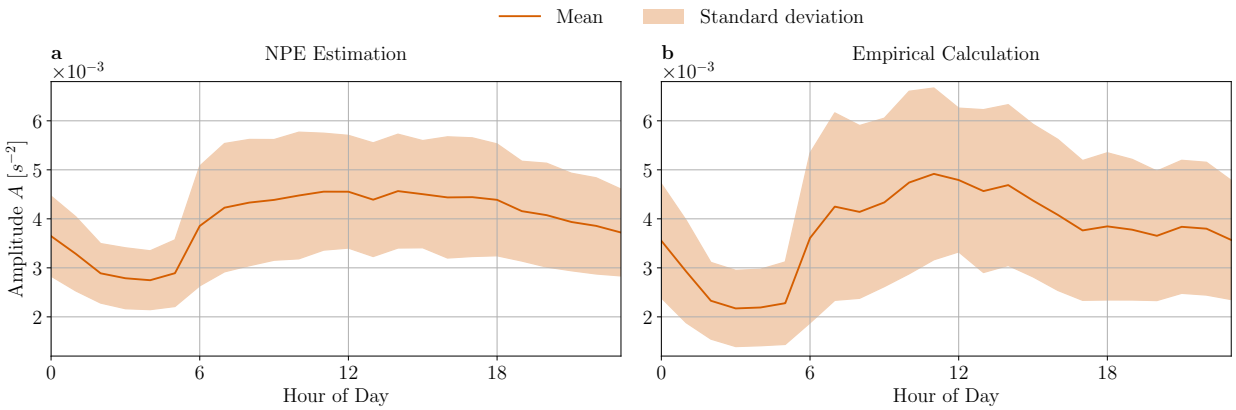
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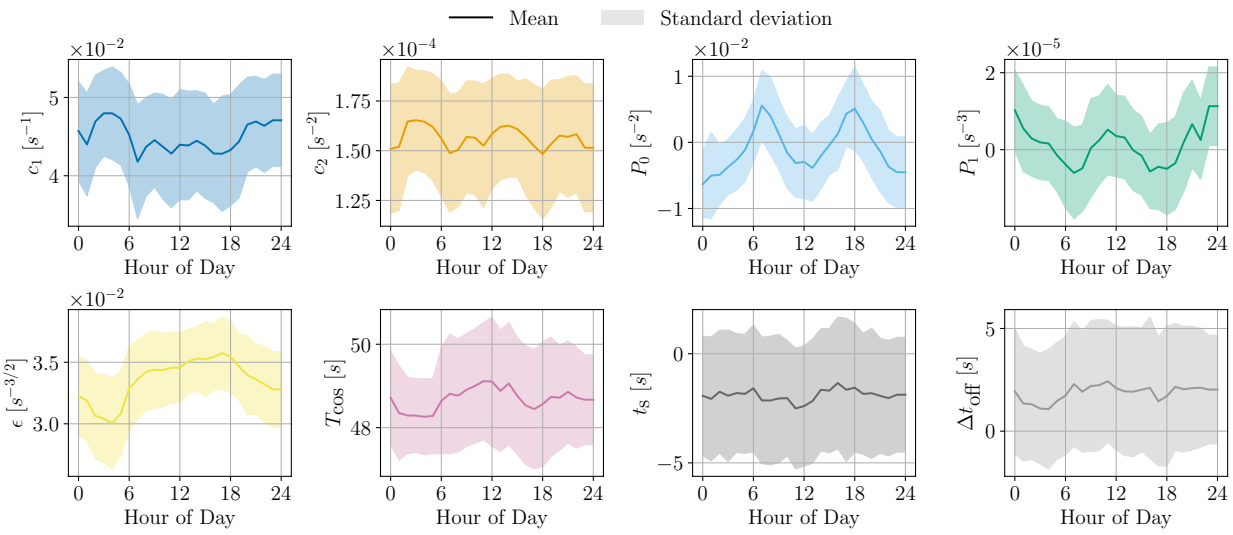
Supplementary Figure 24. The relation of the weekly averaged hourly amplitudes of the 1 min pattern and the amount of power generated by a coal-fired power plant in the Majorcan grid.



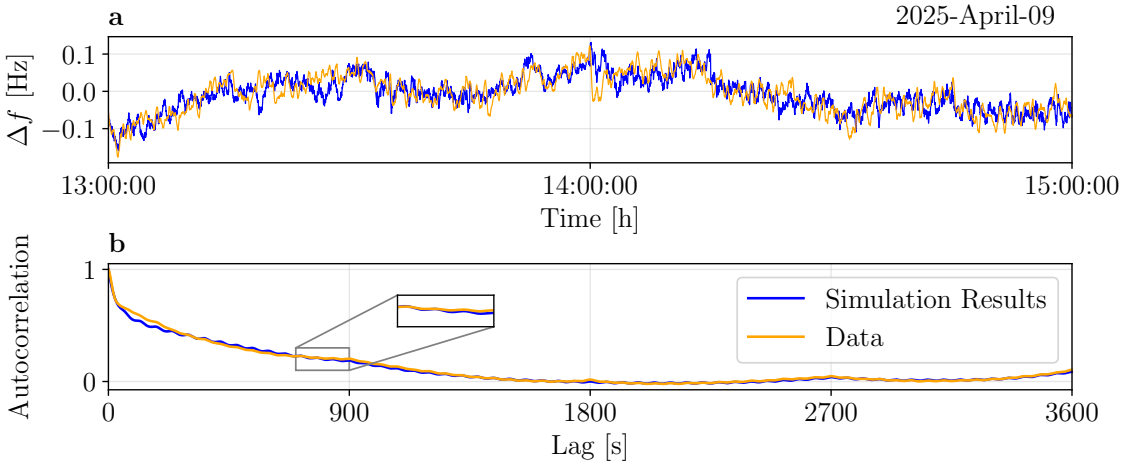
Supplementary Figure 25. Parameter estimation from empirical frequency data via neural posterior estimation (NPE).



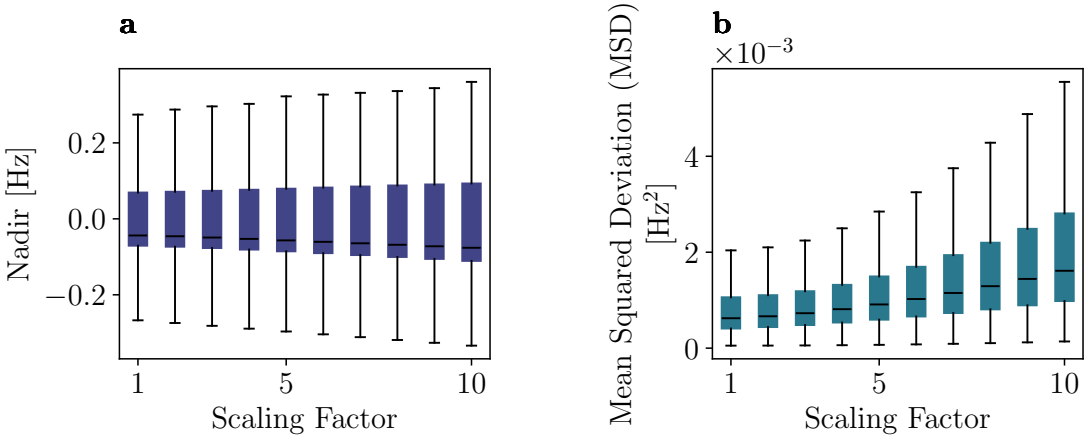
Supplementary Figure 26. **a,** Daily profile of the posterior means of the inferred pattern amplitude. **b,** Daily profile of the empirically calculated amplitude. Solid lines give the mean for each hour of the day, shaded areas the corresponding standard deviation.



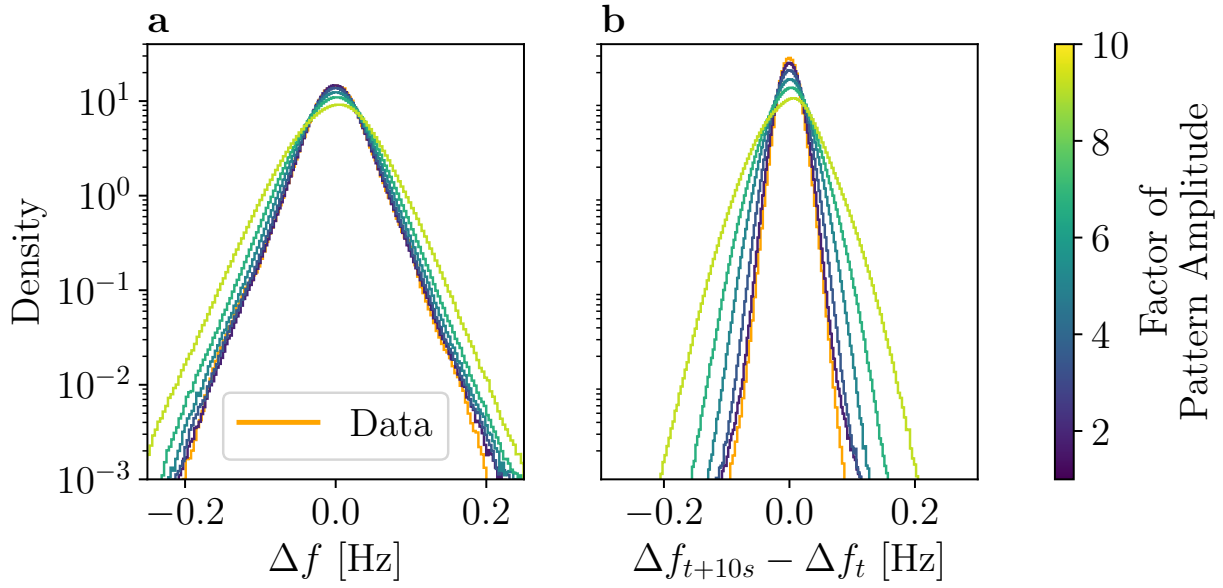
Supplementary Figure 27. Daily profiles of the posterior means for the remaining inferred parameters. Solid lines give the mean for each hour of the day, shaded areas the corresponding standard deviation.



Supplementary Figure 28. **a,** Two-hour segment of the empirical frequency time series of the Scandinavian grid in 2025 together with a trajectory simulated from the NPE parameter estimates. **b,** Autocorrelation of the empirical time series and of the simulations over the same time span.



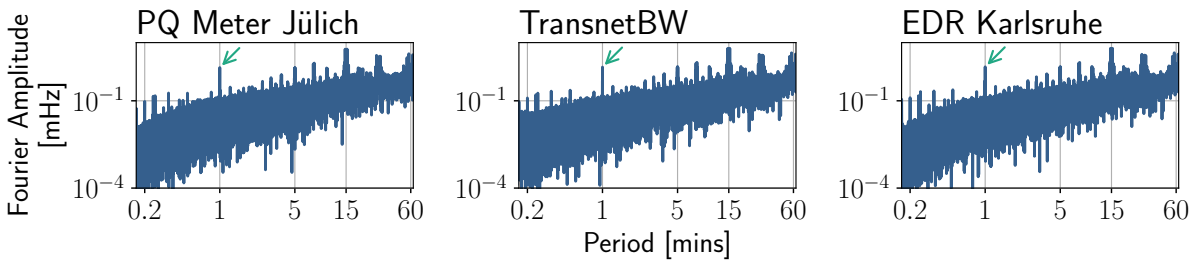
Supplementary Figure 29. Nadir and mean squared deviation of the simulated frequency for increasing amplitude scaling factors of the one-minute disturbance. Box plots show the median (center line), the interquartile range (IQR; box bounds), and whiskers extending to the most extreme data point within 1.5× IQR of the box. Statistics are based on the 27 301 simulated 15 min segments for each value of A.



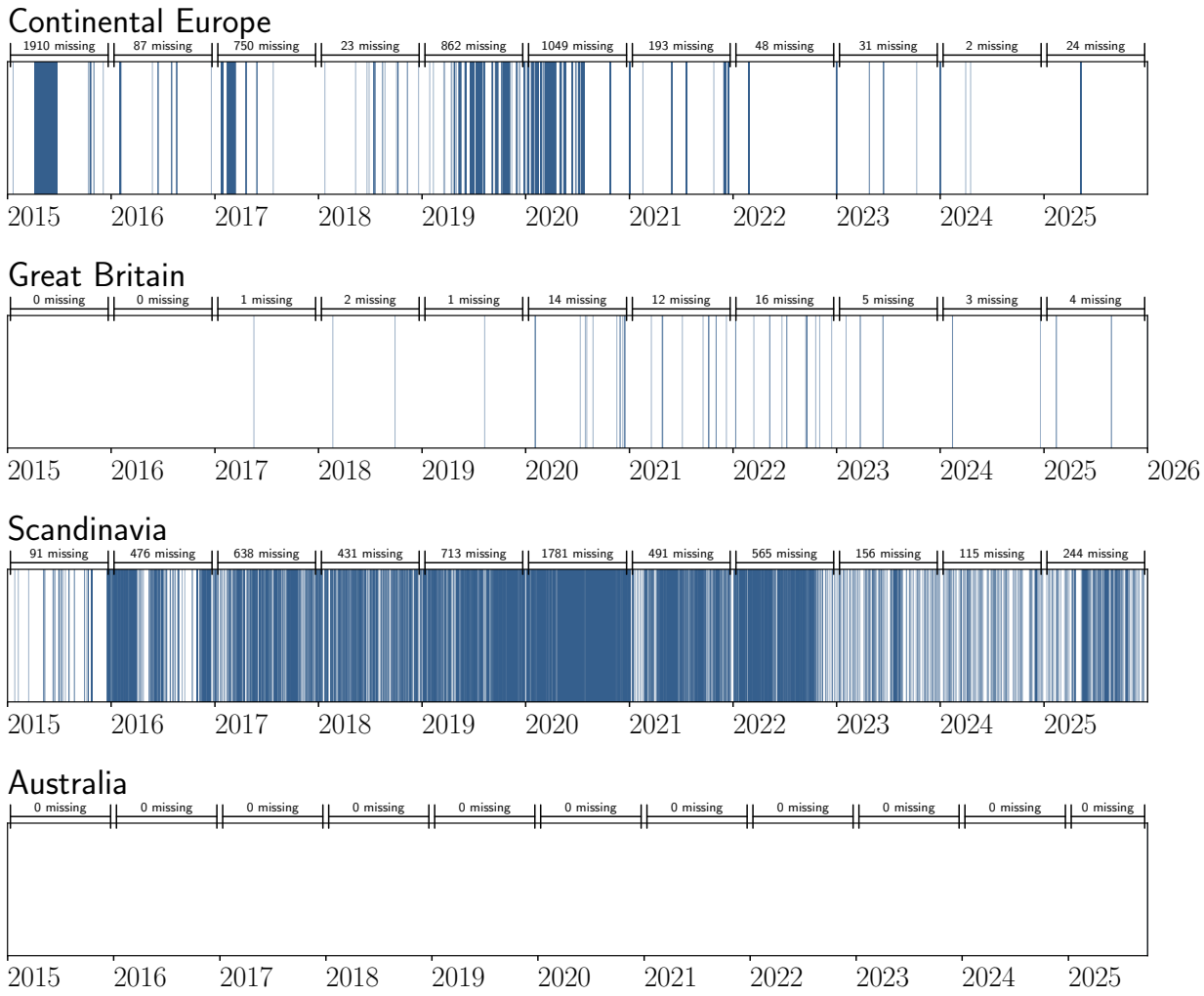
Supplementary Figure 30. **a,** Distribution of the frequency deviations $\Delta f_t = f(t) - f_{\mathrm{ref}}$ for the empirical data set and for simulations with different amplitude factors of the one-minute disturbance. **b,** Corresponding distributions of the 10 s frequency increments $\Delta f_{t+10\mathrm{s}} - \Delta f_t$. The distributions are based on the 27301 segments I_i of 15 minutes from the Scandinavian grid in the time range from 19 March to 31 December 2025.



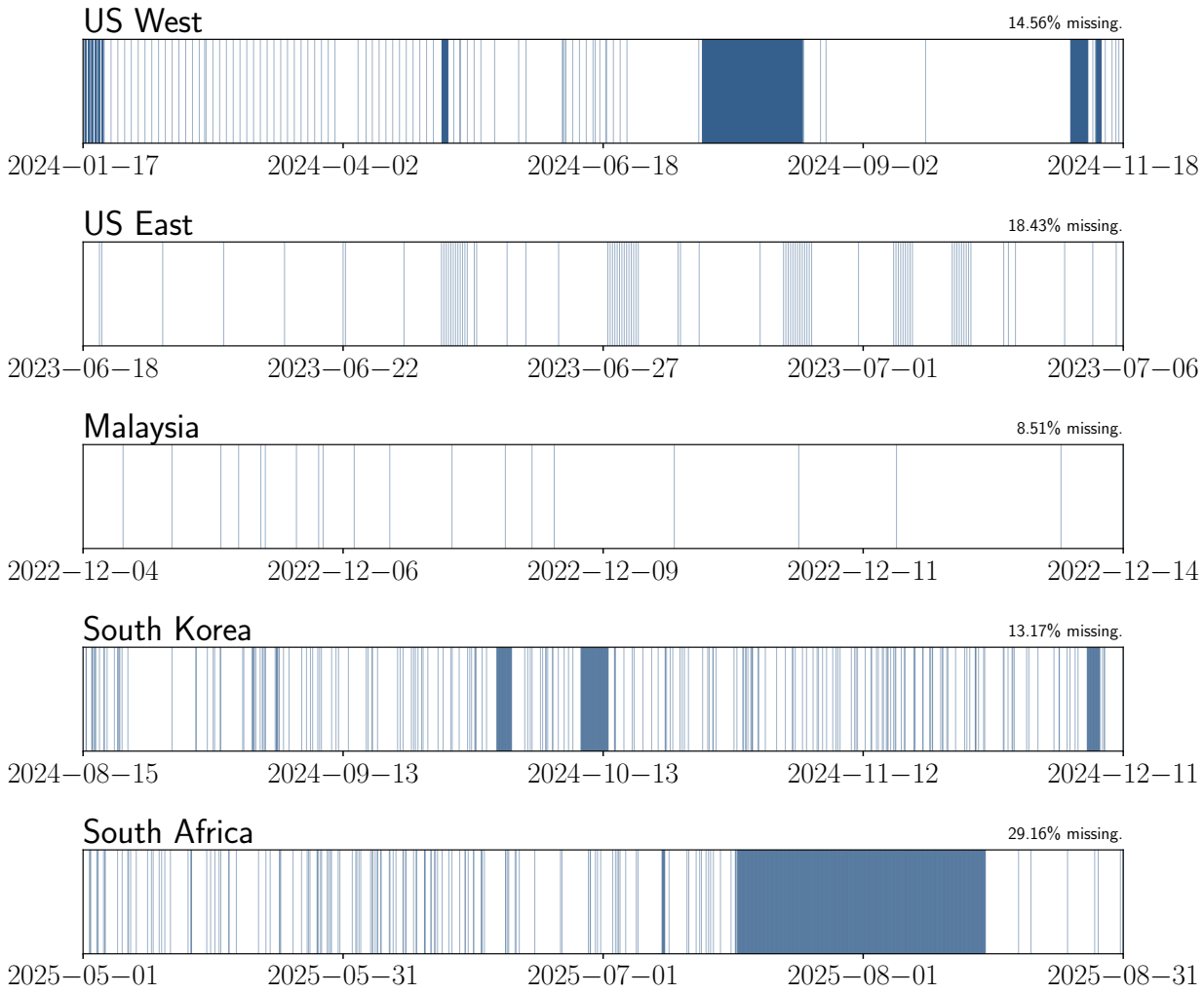
Supplementary Figure 31. EMS software update for Battery Block 1 (cf. electrical diagram in Fig. 5**a**) of the UPS at Building 3. **a**, Comparison of the DC current time series of the three battery blocks before the software update (upper panel) and after the update (lower panel). Before the update, all three battery blocks drew current pulses every minute, synchronized to the same second within each minute. After the update, Battery Block 1 exhibits a more irregular pattern with varying time intervals and longer current pulses, while the overall energy consumption remains approximately constant. Nevertheless, the current draw remains phase-locked to the same second within the minute and is therefore still synchronized with the other two battery blocks, which continue to exhibit a strict 1 min periodicity. **b**, Fourier spectra of the DC currents of the three battery blocks before the update (upper panels) and after the update (lower panels). Following the software update, the sharp spectral peak at 1 min for Battery Block 1 broadens significantly, reflecting the more irregular yet still time-locked operating behavior. In contrast, the other two battery blocks retain a narrow spectral peak at the exact 1 min period.



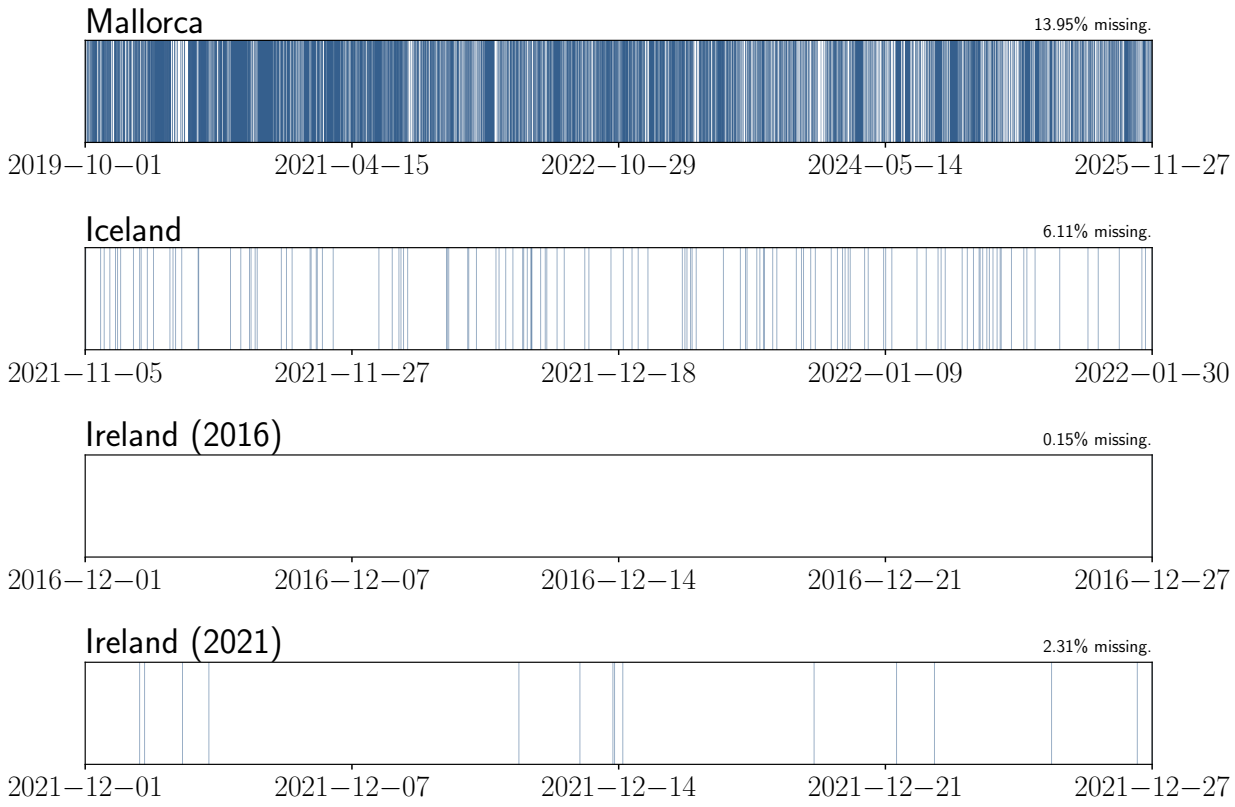
Supplementary Figure 32. Fourier spectrum of frequency measurements from three different sources over a period of 15 days, from June 28, 2025 to July 12, 2025. The time series is given in a time resolution of 3 seconds.



Supplementary Figure 33. Measurement period of the grid frequency $f(t)$ and missing full hours (blue bars) after pre-processing.



Supplementary Figure 34. Measurement period of the grid frequency $f(t)$ and missing full hours (blue bars) after pre-processing.



Supplementary Figure 35. Measurement period of the grid frequency $f(t)$ and missing full hours (blue bars) after pre-processing.

SUPPLEMENTARY NOTE 1: THE 1 min PATTERN IN THE AFRR-SETPOINT IN GERMANY

Automatic frequency restoration reserve (aFRR) corresponds to secondary frequency control in the European power system. The aFRR process aims to restore the Frequency Restoration Control Error (FRCE) to zero and thereby return the system frequency to its nominal value [9]. The corresponding aFRR demand is computed by the transmission system operators (TSOs) responsible for balancing the respective load-frequency-control areas and depends on both frequency deviations and unscheduled power exchanges with neighbouring areas. In practice, the resulting control actions are further modified by imbalance netting within the International Grid Control Cooperation, which reduces simultaneous reserve activations in opposite directions across Europe.

Here, we analyse publicly available aFRR setpoint data for the four German control areas from 2025 at a temporal resolution of 1 s [1]. For simplicity, the setpoints of the four German TSOs are aggregated into a single time series (Supplementary Fig. 3). Fourier analysis reveals a pronounced and highly coherent spectral peak at exactly 1 min. The corresponding waveform is approximately sinusoidal with minor differences between the months of the year. The daily profile of the signal amplitude closely follows the respective curve for the frequency signal amplitude.

The observed oscillation does not directly imply substantial additional activation costs for aFRR because reserve providers are remunerated both for reserved capacity and for activated balancing energy, while positive and negative energy activations approximately compensate over time. However, recurrent periodic changes in power output can contribute to wear and tear of turbines and other assets participating in frequency control [10].

SUPPLEMENTARY NOTE 2: ULTRALOW FREQUENCY OSCILLATIONS AND RESONANCES IN HYDRO POWER PLANTS

Hydro power plants have been identified as contributing factors of slow frequency oscillations [2, 11–13]. These studies suggest that slow periodic oscillations with periods around one minute can arise from resonance effects in turbine–governor dynamics of hydro power plants participating in load-frequency control.

In this work, we adopt the turbine governor model proposed by Odelbrink [2] and investigate its dynamic behaviour. This model allows us to isolate resonance effects and demonstrate that the resonance alone cannot explain the properties of the observed 1 min frequency pattern. Instead, we show that a coherent periodic power imbalance is required to explain the observed long-term stable 1 min pattern.

An overview of the literature on the topic is provided at the end of this Supplementary Note.

A. The Full Hydro Model

1. Frequency Disturbances and Control

We first consider the overall system dynamics as depicted in Fig. 4**a**, where only hydro power plants contribute to the control loop. We start from the aggregated swing equation, which describes the responses of the grid frequency due to power imbalances $\Delta P(t)$ in the grid. Including damping effects, the aggregated swing equation reads [14]

$$M\frac{\mathrm{d}}{\mathrm{d}t}\Delta f(t) + D\Delta f(t) = \Delta P(t), \tag{1}$$

where $\Delta f(t) = f(t) - f_{\mathrm{ref}}$ denotes the frequency deviation from the nominal grid frequency $f_{\mathrm{ref}} = 50\,\mathrm{Hz}$, and $\Delta P(t)$ is the total power imbalance in the grid and $M = 2E_{\mathrm{rot}}/f_{\mathrm{ref}}$ the system inertia. With this definition, $\Delta f(t) > 0$ corresponds to overfrequency and $\Delta f(t) < 0$ to underfrequency. This convention is used consistently throughout this article and facilitates physical interpretation. We remark that the standard control-theory literature, including Odelbrink, often works with the error signal $e(t) = f_{\mathrm{ref}} - f(t) = -\Delta f(t)$ instead. Since the analysis focuses on frequency deviations $\Delta f(t)$, the reference frequency f_{ref} can be omitted.

Taking the Laplace transform of (1) yields

$$(Ms + D)\Delta f(s) = \Delta P(s).$$

Hence, the response of the grid frequency to a power imbalance in Laplace space is given by

$$\Delta f(s) = G_{\mathrm{grid}}(s)\Delta P(s),$$

with the grid transfer function

$$G_{\text{grid}}(s) = \big(Ms + D\big)^{-1}.$$

The total power imbalance $\Delta P(t)$ can be decomposed into external disturbances $\Delta P_{\text{ext}}(t)$ and control actions $\Delta P_{\text{control}}(t)$. External power imbalances can arise due to load fluctuations $\Delta P_{\text{ext}}(t) = -d_L(t)$ or the ramping of power plants. The task of load-frequency control is to compensate for persistent power imbalances and thereby restore the system frequency to its reference value. Combining external disturbances and control actions, the total power imbalance can be written in Laplace space as

$$\Delta P(s) = \Delta P_{\text{ext}}(s) + \Delta P_{\text{control}}(s).$$

We follow Odelbrink and assume that all control power $\Delta P_{\text{control}}(t)$ is provided by hydroelectric power plants consisting of a governor and a turbine; the resulting control structure is summarized by the block diagram in Fig. 4**a**. In particular, the flexible power output of the hydro power plant depends on the valve position $Y(s)$ and is commonly modelled as

$$\Delta P_{\text{control}}(s) = G_{\text{turbine}}(s)Y(s).$$

The governor adapts the valve position $Y(s)$ in response to the frequency deviation $\Delta f(s)$ in order to provide control power

$$Y(s) = G_{\text{governor}}(s)\Delta f(s).$$

The transfer functions for the turbine and the governor are discussed below.

Finally, combining all these expressions and solving for $\Delta f(s)$ yields the closed-loop transfer function

$$\Delta f(s) = \frac{G_{\text{grid}}(s)}{1 - G_{\text{grid}}(s)G_{\text{turbine}}(s)G_{\text{governor}}(s)}\Delta P_{\text{ext}}(s), \tag{2}$$

which models the frequency response due to external power imbalances $\Delta P_{\text{ext}}(s)$ in the full grid-governor-turbine control system.

2. Turbine and Governor Model

Odelbrink defines the turbine transfer function as

$$G_{\text{turbine}}(s) = \frac{-T_w s + 1}{T_w s/2 + 1}.$$

This transfer function is widely used to represent systems with transport and storage effects. It captures the finite time T_w required for the turbine to accelerate after a valve opening.

Odelbrink proposes a governor model consisting of the following components:

- a proportional–integral (PI) controller,
- a delay of duration T_{del}, representing mechanical and hydraulic delays within the plant,
- a droop control, introducing a feedback loop with gain EP, where EP is the inverse regulation strength in percent of rated power,
- an attenuation factor $c \in [0, 1]$, accounting for the fact that only a fraction of the nominal power is available for load-frequency control.

The block diagram of this control system, cf. Fig. 4**b**, can be translated into equations as follows. The input to the PI controller is given by

$$-\Delta f(s) - EP \cdot \tilde{Y}(s),$$

where $\tilde{Y}(s)$ denotes the valve position before attenuation. Multiplication with the PI controller transfer function (and the delay element) yields

$$\tilde{Y}(s) = e^{-sT_{\text{del}}} \left(K_P + \frac{K_I}{s} \right) \left(-\Delta f(s) - EP \cdot \tilde{Y}(s) \right).$$

Solving for $\tilde{Y}(s)$ and including the attenuation factor c gives

$$Y(s) = G_{\text{governor}}(s) \Delta f(s),$$

with the governor transfer function

$$G_{\text{governor}}(s) = -c \frac{e^{-sT_{\text{del}}} \left(K_P + \frac{K_I}{s} \right)}{1 + EP \cdot e^{-sT_{\text{del}}} \left(K_P + \frac{K_I}{s} \right)}.$$

3. Parameters

We take the parameters from Odelbrink which are summarized in the following Table.

D	K_P	K_I	EP	c	M	T_w	T_{del}
1 (0.1)	2	1/0.6	0.01	0.11	18	1.25	0.3

For the damping coefficient, Odelbrink suggests $D = 1$. In this work, we also investigate a system with weaker damping $D = 0.1$ to evaluate stronger resonance effects.

B. The Extended Model

The extended model incorporates a heterogeneous provision of load-frequency control by allowing only a fraction $h \in [0, 1]$ of the control action to be supplied by hydro power plants, while the remaining share $1-h$ is provided by a generic proportional–integral (PI) controller with transfer function

$$G_{\mathrm{PI}}(s) = -\left(K_P + \frac{K_I}{s}\right),$$

which directly acts on the frequency deviation without including turbine or delay dynamics.

Accordingly, the total control action is modelled as

$$G_{\mathrm{control}}(s) = h\, G_{\mathrm{turbine}}(s) G_{\mathrm{governor}}(s) + (1-h)\, G_{\mathrm{PI}}(s).$$

Substituting this expression into the closed-loop dynamics yields the extended transfer function

$$\Delta f(s) = \frac{G_{\mathrm{grid}}(s)}{1 - G_{\mathrm{grid}}(s)\, G_{\mathrm{control}}(s)} \Delta P_{\mathrm{ext}}(s). \tag{3}$$

C. Numerical Results

1. *Frequency Response at a period of* 60 s

Figure 5 shows the Bode diagram of the full hydro closed-loop system (2). A pronounced resonance peak at a period of approximately 60 s is clearly visible, whose magnitude depends on the damping parameter D. In addition, the resonance behavior is strongly influenced by the attenuation factor c. The largest frequency response at the 60 s period is observed around $c = 0.11$, see Fig. 6**a**. This behavior can be understood by inspecting the Bode diagrams for different values of c, shown in Fig. 6**b**.

For $c = 0$, no load-frequency control is active, and low-frequency disturbances are not suppressed, leading to large and persistent frequency deviations. As c increases, low-frequency disturbances are increasingly damped. However, this comes at the cost of an emerging feedback effect that leads to the resonance peak with a period on the order of 60 s. Moreover, the location of this peak shifts towards higher frequencies as c increases, illustrating how the strength and position of the resonance depend on the amount of hydro power reserved for load-frequency control.

The extended system (3) allows us to continuously interpolate between the pure hydro system ($h = 1$) and a fully synthetic PI-controlled system ($h = 0$). Increasing the hydro share h leads to a monotonic increase in the frequency response at the 60 s period, see Fig. 7**a**. This trend can likewise be explained by the corresponding Bode diagrams, see Fig. 7**b**. The generic PI controller exhibits a weaker resonance at higher frequencies and does not reproduce the pronounced peak observed in the hydro-based system.

Although the generic PI controller is a simplified representation and does not fully capture the dynamics of other power plants participating in load-frequency control, it nevertheless illustrates how the dispatch of hydro power influences resonance phenomena. In particular, the amplitude of the system response to disturbances with a characteristic period of 60 s depends sensitively on the composition of control resources, especially on the share of hydro units contributing to frequency control.

2. Time Domain Simulations

We next simulate the frequency response in the time domain under two scenarios for the external power imbalance $\Delta P_{\text{ext}}(t)$. We show results for $D = 1$ and $D = 0.1$, where the latter value corresponds to an artificially weak damping. This parameter is chosen for illustrative purposes, as resonance effects are particularly pronounced.

First, we consider a power imbalance $\Delta P_{\text{ext}}(t)$ that leads to a typical deterministic frequency deviation (DFD). We assume that generation is strongly ramped at the beginning of 15 min market interval leading to a rapid increase or decrease of the power imbalance $\Delta P_{\text{ext}}(t)$, which gradually returns to zero afterwards (cf. Fig. 8**a**). Second, we consider the same overall power imbalance $\Delta P_{\text{ext}}(t)$, but add a small periodic modulation with a period of 60 s, representing a periodic power imbalance due to clocked devices (dashed line in Fig. 8**a**)

In the absence of the 60 s modulation, a transient 60 s resonance oscillation is excited, but it is rapidly damped. Moreover, the oscillations generated during ramp-up and ramp-down DFDs are phase-shifted by 180°. As a consequence, such resonance-induced oscillations cancel out when averaged over multiple market intervals and therefore (1) do not appear as a stable peak in long-term Fourier spectra and (2) are not consistent with the actual frequency time series, cf. the daily profile in the Nordic grid (Fig. 1).

In contrast, when a periodic 60 s power imbalance is present, a persistent and phase-coherent 60 s frequency pattern emerges. Crucially, towards the end of the market interval no phase shift is observed between the oscillations generated by initial ramp-up and ramp-down of the power imbalance. This demonstrates that a weak external periodic driver can introduce a stable, coherent 60 s oscillation of the grid frequency, while a pure resonance effect cannot explain the stability of the signal amplitude and phase.

Finally, we also add noise to the power imbalances $\Delta P_{\text{ext}}(t)$ (Fig. 9). We first consider the scenario without the 60 s modulation. Interestingly, we observe a transient ~ 60 s pattern in the frequency response on short time scales. However, these patterns are *not* phase coherent. Evaluating the response to a sequence of DFD inducing power ramps, the Fourier spectrum shows no distinct peak at a period of 60 s in contrast to the observations. In contrast, when a small signal with a period of 60 s is added to $\Delta P_{\text{ext}}(t)$, a clear and stable peak at a period of 60 s emerges in the Fourier spectrum.

In conclusion, resonance effects alone cannot explain the observed 60 s patterns observed

in grid frequency time series worldwide. Without an external driver, oscillations lack phase-coherence and long-time stability. In contrast, a weak periodic and phase-coherent component in the power imbalance readily explains the observed stable 60 s pattern in the grid frequency. Resonance effects can amplify this signal depending on the design and parameters of the control system.

D. Literature Overview

In the scientific literature, the phenomenon of ultra or very low frequency oscillations repeatedly appears in conjunction with hydro-dominated power systems. We found the following resources to be especially helpful to understand the scope and potential impacts of the phenomenon across different power grids.

Reference	Summary
[13]	Analyses large amplitude (< 1 Hz) resonances in the hydro-dominated Columbian power system, focus on its dominant 20 s oscillation mode.
[11]	Analyses the influence of hydro-turbine governor settings on oscillations with a period of 1 min in the hydro-dominated Nordic power system.
[2]	Develops a model of a hydro-dominated power system leading to oscillations with a period of 1 min and suggests an alternative control design.
[15]	Provides measurements and simulations of oscillations with a period of 1 min in the Nordic power grid, including a 50 MW excitation experiment.
[10]	Analyzes mechanical wear in control elements of hydro power plants via valve travel and impact of 1 min oscillations in the Nordic power grid.
[16]	Provides a long-term (2015 - 2021) analysis of ULFOs in the Nordic power system.
[17]	Analyzes the relation of ULFOs and system inertia, showing that signal amplitudes increase with decreasing inertia.
[18]	Proposes a damping control index for ULFO suppression and validates it with data from the Southwest China power grid.
[19]	Identifies that negative damping from hydraulic governors causes ULFO while the exciter and Power System Stabilizer (PSS) have minimal impact.
[20]	Develops a hydro-turbine governor additional damper (GAD) based on phase compensation to eliminate negative damping.
[21]	Introduces a distributed governor control design to ensure frequency stability in multi-machine systems with high renewable integration.
[22]	Analyzes how integrating wind power into hydro-dominant systems provides positive damping to suppress ULFO.
[23]	Develops a dynamic model for a hydro power plant with focus on the oscillatory behavior in the British power grid with a period of 20 s.

SUPPLEMENTARY NOTE 3: GBT MODEL FOR THE HOURLY AMPLITUDES IN THE CE GRID

We investigate the role of potential contributing factors to the 1 min pattern using methods from eXplainable Artificial Intelligence (XAI) [24, 25]. We train and analyze Gradient Boosted Tree (GBT) models to predict the hourly amplitudes pattern in the CE grid. GBTs are lightweight to train [26], provide state-of-the-art performance for tabular data [27], and offer intrinsic interpretability [28].

Two models are considered, each using a different feature set. The *actual* model includes hourly aggregated load and generation by technology type (e.g., solar, gas, nuclear), as well as derived features such as generation ramps, defined as the difference in generation between two consecutive hours. The *full* model extends this feature set by incorporating day-ahead forecasts and day-ahead prices, together with additional derived features, including forecast errors and artificial temporal encodings for hour of day, weekday, and month. The data for the techno-economic features were taken from the ENTSO-E transparency platform [29], following the procedure outlined in [30].

Model predictions are interpreted using SHapley Additive exPlanations (SHAP) [28], computed exclusively on the test set. This approach has previously been applied successfully to explain DFDs and control activation mechanisms [30, 31].

The results (cf. Fig. 10) show that hourly amplitudes can be partially predicted from techno-economic features. Both models achieve similar predictive performance ($R^2 = 0.46$ for the actual model and $R^2 = 0.47$ for the full model) and outperform the daily profile benchmark, which is based on historical information, on the unseen test set (panel **a**). However, the feature importance ranking of the full model, computed using mean absolute SHAP values, reveals that the daily profile remains largely unexplained, as the hour of day is the dominant feature. This suggests that the actual model instead reconstructs the hourly profile through a combination of multiple techno-economic variables (panel **b**). Nevertheless, both models consistently identify hydro, solar, nuclear, and gas generation among the most influential features.

SHAP dependence plots for the full model (panel **c**) indicate that higher renewable generation, particularly solar, pumped hydro, and reservoir hydro, is associated with stronger amplitudes, reflected by positive SHAP values. In contrast, conventional generation sources such as nuclear and gas exhibit a negative correlation between generation levels and SHAP values, indicating a damping effect on the amplitudes. The findings for solar, nuclear, and gas are consistent with the role of inertia in damping low-frequency oscillations.

The positive correlation observed for hydro generation suggests that resonance effects, as discussed in the main text, contribute to the signal. We find that SHAP values for both pumped hydro and reservoir hydro generation are on the order of ± 0.1 mHz. Most interestingly, the partial dependency plot for pumped hydro generation shows a sharp decrease when the feature value drops below 2 GW while it is largely constant above this value. This

can be interpreted as a step-wise contribution to the signal amplitude, with negative SHAP values if most turbines are offline and positive SHAP values when most turbines are online but independent of the actual generation.

The results of our data-driven analysis thus lead to the conclusion that hydro power contributes to the signal amplitude in the CE grid, albeit at a moderate scale of $\pm 0.1\,\mathrm{mHz}$. Given the results of the mechanistic model, hydro power should be viewed as an amplifying factor rather than the cause of the observed pattern.

SUPPLEMENTARY NOTE 4: THE IMPACT OF COVID-19 ON THE 1 min PATTERN IN THE CE GRID

The COVID-19 pandemic had a strong impact on the economic activities and daily life across the globe and thus also affected the way electric power is consumed. The pandemic reached Europe in early 2020 and many countries issued a lockdown to suppress new infections in spring 2020. Figure 11 illustrates two characteristic aspects of the lockdown in Germany in April 2020. Industry production decreased substantially, in particular in the automotive sector. At the same time the internet traffic increased strongly, for instance, due to extended home office or video streaming.

The weekly amplitudes of the 1 min pattern increased strongly by a factor of $1.3 - 1.4$ at the beginning of the lockdown. There is no corresponding decrease at the end of the lockdown, as this date coincides with the seasonal maximum discussed above. The amplitude remains high and decreases only later during the year.

We conclude that it is unlikely that industry processes are the main cause of the 1 min pattern. During the lockdown, industrial activities decreased strongly, while the amplitude of the 1 min increased. In contrast, industry processes may contribute to the damping of frequency variations by providing inertia.

The utilization of information and internet infrastructures, measured by the internet traffic at the DE-CIX internet node, shows a strong increase during the lockdown, just as the pattern amplitude. Large computing centres typically operate uninterruptible power supplies (UPSs) with BESSs to avoid data loss in the case of a power outage. Hence, it appears possible that increased internet traffic correlates with a stronger use of UPSs.

SUPPLEMENTARY NOTE 5: REVISION OF THE PRIMARY CONTROL REQUIREMENTS IN THE SCANDINAVIAN GRID

In this section, we discuss the revision of the Frequency Containment Reserve (FCR) requirements in the Nordic synchronous area, comprising Norway, Sweden, Finland, and eastern Denmark. In the main text, we use the term Scandinavian power grid for readability, although the formally correct designation used by ENTSO-E is Nordic synchronous area.

Around 2010, the amplitude of the coherent ultra-low-frequency oscillation (ULFO) mode in the Scandinavian grid regularly exceeded 10 mHz. However, the amplitudes reported in this work are defined for the coherent waveform reconstructed over individual one-hour intervals and therefore characterize only the long-term coherent component of the oscillation. Larger transient deviations may occur during shorter oscillatory events or following disturbances [17].

At the same time, increasing frequency fluctuations and decreasing system inertia became a growing operational concern for the Scandinavian transmission system operators (TSOs). The Nordic power system has long been known to exhibit poorly damped low-frequency oscillatory behaviour, and a substantial body of literature discusses resonance phenomena in hydroelectric power plants and the associated mechanical wear of hydraulic turbines, waterways, and governor systems (cf. the literature review above). Against this background, the Scandinavian TSOs initiated a major revision of the Frequency Containment Reserve (FCR) framework and corresponding prequalification procedures. According to the explanatory documents accompanying the revised methodology, the previously implemented FCR requirements had been designed for a system with substantially higher inertia, whereas lower inertia conditions increase the need for fast and well-damped frequency control responses. The revised requirements therefore explicitly aimed to improve the dynamic performance of FCR provision and to ensure sufficient damping of slow frequency oscillations with characteristic periods around one minute.

The revised technical requirements, implemented in 2023, introduced substantially stricter prequalification criteria for participating FCR providers [32]. In particular, the new framework added explicit requirements concerning dynamic performance, linearity, and oscillatory stability. Providers must now pass standardized sine-response tests and dynamic ramp tests designed to evaluate the response of the reserve-providing units over oscillatory timescales including periods around one minute. The revised rules furthermore require that FCR responses contribute positively to the stabilization and damping of system frequency oscillations. Additional linearity requirements were introduced to reduce dead bands, backlash, and nonlinear response characteristics that can otherwise amplify oscillatory behaviour. The revised framework also distinguishes between dynamic and static variants of disturbance reserves (FCR-D), ensuring that a sufficiently large fraction of reserves is capable of continuously following rapid frequency variations under low-inertia conditions. Overall, the revised Scandinavian FCR framework can therefore be interpreted as a direct operational

response to the increasing importance of low-frequency oscillatory dynamics in the power system.

The technical requirements for Frequency Containment Reserves (FCR) introduced in 2023 were revised in 2025 to further expand standardized dynamic performance testing of reserve-providing units [33]. The revised requirements explicitly include sine-response tests with oscillation periods of 50–70 seconds, i.e. on approximately one-minute timescales, and specify detailed procedures for gain and phase characterization under different loading and droop conditions.

SUPPLEMENTARY NOTE 6: GEOGRAPHICAL ANALYSIS OF THE 1 min PATTERN IN THE *CE* GRID

We now compare frequency recordings from various locations in the *CE* grid for one week in July 2019, which may provide hints towards the causes of the 1 min pattern. For example, if the source was localized, we expect the pattern to be strongest in the vicinity of the source. We thus extract the amplitudes and average waveforms of the pattern from synchronized measurements for all locations, as these are our main statistical indicators for the 1 min pattern.

Most recordings have been performed on ordinary pocket sockets on the distribution grid level, among those are the recordings obtained by scientists of the Karlsruhe Institute of Technology (KIT) in Lisbon (Portugal), Karlsruhe, Oldenburg (Germany) and Istanbul (Turkey) [6] and by the company *GridRadar* in various locations mostly in Germany [5], cf. map in Fig. 12. We compare these frequency recordings with measurements on the transmission grid level performed by German transmission system operators (TSOs) [1]. TSO data has been provided by TransnetBW localized in Stuttgart before moving to the new common data platform [1]. The measurement period covers one week in July 2019, but the KIT dataset is missing some recordings for some hours. We exclude these hours when we compare the locations between different datasets.

The 1 min pattern shows some interesting regional differences. The average hourly amplitudes are generally stronger in the periphery of the grid, that is in Istanbul and Lisbon, compared to the core in Germany, cf. Fig. 12a. This is consistent with earlier studies [34] on spatial features of frequency fluctuations. Fluctuations in the outskirts of the *CE* grid are generally stronger than in the centre. However, measurements in Schaefstall in Bavaria, Germany, break this rule. The average amplitudes are significantly stronger compared to the other locations in the vicinity, cf. Fig. 12a. To further investigate this 'Schaefstall Anomaly' we plot the average amplitudes over the course of a day, cf. Fig. 12b. The daily profiles reveal that the anomalous high average amplitude in Schaefstall stems from high amplitudes at certain hours of the day, in particular 4 am, 4 pm and 9 pm. Furthermore, the amplitudes in Schaefstall are (almost) constantly higher than at the other locations for these hours, cf. Fig. 13. This suggests that the anomaly is not caused by a faulty measurement by the recorder in Schaefstall and thus might give a hint to potential local drivers or amplifying factors due to a resonant behaviour.

To identify potential local contributing factors, we investigate which generators and consumers are in the vicinity of the frequency recorder in Schaefstall, see Fig. 14. There are two large factories, a sugar factory in Rain and an Airbus Helicopters factory in Donauwoerth, that are connected to the local distribution grid according to [7]. Each has its own gas power plant. Furthermore, we find several run-of-river hydro plants on the side rivers of the Danube. These, among others, could be potential local driving or amplifying factors. We recall that hydro power plants have been identified as contributing factors of slow frequency

oscillations as discussed in Supplementary Note 2 and Refs. [2, 11–13].

Finally, we also consider the respective local waveforms, which show slight differences in the phase shift, increasing from West to East (Fig. 12). However, all measurements have been performed on ordinary pocket sockets in the low-voltage distribution grid, impeding the interpretation of the results. Indeed, a much stronger difference is observed between recordings from the distribution grid in Karlsruhe and from the German TSOs in close geographic proximity. Remarkably, the pattern extracted from the TSO data is lagging behind. This may indicate that relevant driving or synchronization processes originate predominantly in the distribution grid with the transmission grid being driven and following with a certain lag.

SUPPLEMENTARY NOTE 7: POWER MEASUREMENTS IN JUELICH

This section presents the analysis conducted on the data obtained from the FZJ campus electrical grid. After a brief overview of the electrical system and the measurement infrastructure, the Fourier spectra of active and reactive power for selected significant locations are provided.

Forschungszentrum Jülich Electrical Grid

The electrical grid of the Jülich research campus consists of multiple medium voltage (MV) distribution feeders at 10 kV, the majority of which are reclosing on the same busbar at the primary substation. This configuration forms ring topologies, which are typically employed to increase the resiliency of the grid. At the different nodes of the feeders, the medium voltage (MV, 10 kV) to low voltage (LV, 0.4 kV) transformers connect to office and laboratory buildings, which could also be equipped with a UPS including a BESS or photovoltaic power sources. Moreover, the electrical grid of the campus is characterized by large-scale distributed energy resources, including a large photovoltaic power plant. Fig. 15 provides a schematic representation of a portion of one of the campus's distribution rings where buildings are connected to the MV level via transformers.

Measurement Infrastructure of the Campus

The electrical grid of the FZJ campus is continuously monitored at multiple locations, with over 80 PQI-DA smart power quality (PQ) meters manufactured by the company a-eberle [35]. These devices, which are in compliance with IEC 61000-4-30 Ed.3 (2015) standard for an A-Class device [36], are capable of providing synchronized measurements of several electrical quantities with an accuracy of 0.1% at the fundamental frequency. It is worth noting that the monitoring devices used in the measurement campaign were recently purchased and are commercial, state-of-the-art power quality analyzers. The manufacturer has certified these instruments to comply with all major international power quality standards, including EN 50160, IEEE 519, IEC 61000-2-2, IEC 61000-2-4, IEC 61000-2-12, IEC 61000-4-7, IEC 61000-4-15 and IEC 61000-4-30, as well as NRS 048. Compliance with these standards is verified and certified prior to delivery, ensuring that the devices meet the required accuracy, measurement methods and performance criteria for power quality assessment. Therefore, the validity and reliability of the recorded measurements used for this analysis are ensured by the certified compliance of the monitoring equipment with internationally recognised standards. By means of the Modbus TCP protocol, different registers of the PQ meters are used to collect voltage, current, phase angle and frequency from different locations of the campus. The timestamped and synchronized measurements are acquired

with a 3-second time interval and stored on the campus FIWARE-based ICT platform [37], which allows for a smooth extrapolation of the data by selecting the time period and data resolution.

As previously explained, the campus electrical grid is composed of several main MV distribution feeders, to which one or multiple buildings are connected by means of MV/LV transformers. Some large buildings use multiple transformers to supply different portions of the building due to high load demands. An explanatory description of the 10 kV MV distribution grid of the campus and the connection to the 400 V LV system is presented in Fig. 15, which shows several MV/LV transformers connected to the same MV bus to supply one or more buildings. For this reason, the data from multiple PQ meters has been used to characterize the power quality and calculate the Fourier spectra for some of the campus' buildings.

For this analysis, the data of two different periods have been considered:

- MP1: from 2024-03-24 00:00:00 to 2024-07-03 23:59:57
- MP2: from 2024-10-01 00:00:00 to 2024-11-05 23:59:54

In addition, the data collected from a high-power BESS (1.5 MVA / 0.5 MWh) installed in the campus has been considered. The system has a rated voltage of 400 V (480 Vdc) and a maximum load current of 2.5 kA. The BESS is permanently connected via a dedicated transformer to the electrical grid, and it is typically operated as a UPS. As described in the main manuscript, the UPS topology exhibits three battery racks connected in parallel to the DC link between the grid-side converter and the building-side converter.

Result of the Data Analysis of the Campus

For each building of the campus where PQ meters are installed, the measurements of active and reactive power from all the PQ meters associated with the same building have been collected for both time periods, MP1 and MP2. Then, the Fourier spectra have been calculated for each set of measurements, as displayed for the most significant ones in Fig. 16 and Fig. 17 for active and reactive power, respectively. We evaluate if the spectra show a peak at 1 min or other significant patterns. The findings are summarized in Table 1.

PQ ID	Building	Data		1 min Pattern		Description
		MP1	MP2	P	Q	
002	1	✓	✓	-	-	High Performance Computer
003	1	✓	✓	-	-	High Performance Computer
004	1	✓	✓	MP1	both	UPS, Offices
005	1	✓	✓	MP1	both	UPS, Offices
006	1	✓	✓	both	-	UPS, Offices
007	1	✓	✓	both	-	UPS, Offices
014	2	✓	-	-	MP1	Offices
015	2	✓	-	-	MP1	Offices
023	3	✓	✓	both	both	UPS (1.5 MVA / 0.5 MWh at 400 V)
051	4	✓	✓	both	MP1	UPS, Offices, PV, BEV charging
083	5	✓	✓	-	both	PV field
084	5	✓	✓	-	both	PV field
088	6	-	✓	MP2	-	UPS, Offices

Supplementary Table 1. Overview of all PQ Meters with interesting patterns in the active P (cf. Fig. 16) or reactive power Q (cf. Fig. 17). We compare two measurement periods, MP1 and MP2. The data columns indicate for which PQ Meter we collected data in the two periods.

The results indicate that the Fourier spectrum for the PQ meter 023 has a very sharp peak at 1 minute for both active and reactive power in both data sets. This assertion is supported by the fact that the measurement is collected after the MV/LV transformer that connects the high-power BESS (1.5 MVA / 0.5 MWh) that currently serves as a UPS. This device, as described in the main paper, injects current peaks with a 1 minute period into the grid, thereby generating the distinct peak in the spectrum.

Similarly, the same type of peaks are visible in measurement sets of other buildings. Specifically, it can be noticed that the PQ meters 006, 007, 051, 088 show a distinct peak at 1 minute for active power. Notably, PQ meter 088 was online only in measurement period MP2. These four PQ meters monitor the transformers supplying different buildings that all host one or multiple UPSs based on BESSs.

The PQ meters 004 and 005 show small peaks at 1 minute for active power during the measurement period MP1. These peaks are hardly visible on a logarithmic scale as in Fig. 16, but clearly visible on a linear scale. We do not observe comparable peaks during the measurement period MP2, though. These two PQ meters monitor the transformers

supplying one building, which hosts one UPS based on a BESS.

In addition, we observe broad peaks in the Fourier spectrum of the active power consumption at approximately 70 seconds for the PQ meters 014 and 015 supplying typical office buildings. The strong dispersion of the peaks points towards a recurrent pattern that is either short in duration, strongly damped or subject to noise or jitter. In any case, this signature strongly differs from the stable periodic patterns observed for other PQ meters at 1 minute.

Finally, the Fourier spectrum of the reactive power time series shows a peak at 1 minute for several PQ meters. These PQ meters serve different types of buildings as well as photovoltaic power sources. These patterns deserve further investigation.

SUPPLEMENTARY NOTE 8: BACKGROUND ON BATTERY ENERGY MANAGEMENT SYSTEMS

Regulatory requirements and market conventions shape the time scales on which battery energy management systems (EMS), aggregators, and virtual power plants (VPPs) interact with the grid. In European FCR frameworks, relevant response requirements operate on time scales from seconds to tens of seconds. In Continental Europe, full FCR activation is conventionally associated with a 30 s time scale, while Nordic FCR specifications define product-specific dynamic-performance, measurement, and verification requirements over several time scales [32, 38]. These requirements constrain EMS and inverter-control design, because resources must respond rapidly to frequency deviations while avoiding unstable control action, unnecessary cycling, and excessive degradation.

Some deterministic imprints in power-system frequency are well documented. ENTSO-E reports deterministic frequency deviations in European frequency data and discusses them mainly in relation to market schedules, dispatch changes, and ramping behaviour [39]. Kruse et al. further show that power-grid frequency contains systematic, partly predictable structures, indicating that frequency trajectories are influenced by recurring operational patterns [40]. These findings demonstrate that synchronized operational processes can become visible in aggregate frequency measurements.

At the low-level control and measurement layer, FCR-providing resources require comparatively fast frequency measurement and response capabilities. NREL studies show that distributed inverter-based resources, including storage inverters, can provide fast frequency support through functions such as frequency-watt control [41]. In contrast, higher-level EMS functions such as SOC management, recharge scheduling, ramp-rate constraints, and aggregator dispatch are implementation-dependent and may operate at slower, discrete update intervals.

Therefore, recurring minute-scale components in frequency measurements should be interpreted as potential signatures of synchronized control rather than as generic measurement artifacts. In the context of the present study, UPS battery systems emerge as a plausible contributor to such periodic forcing. One possible mechanism is periodic diagnostic or health-management activity within UPS and battery EMS, including self-tests, state-of-health checks, recharge control, or brief supervisory adjustments of charging and discharging states.

Such minute-scale diagnostic routines are explicitly documented by multiple commercial UPS manufacturers. For example, one manufacturer states that the battery circuit is tested automatically every 60 s to detect open circuits or increased resistance [42]. Another specifies that battery testing is performed automatically every 60 s for basic battery diagnostics, complemented by a battery-efficiency test every 220 min during float charging [43]. Similar periodic battery-connection self-tests have also been described for other commercial UPS systems [44]. Although these documents describe individual products rather than the entire

UPS market, they demonstrate that minute-scale diagnostic routines are implemented by multiple independent vendors and therefore represent a plausible physical mechanism for synchronized minute-scale power perturbations when executed across large populations of devices.

While the power perturbation of an individual UPS system is small, similar EMS routines executed across large fleets of devices could aggregate into measurable frequency-domain features if their timing is synchronized by firmware defaults, communication cycles, clock-based scheduling, or aggregator-level coordination. This makes UPS battery EMS timing a credible mechanism for shaping minute-scale grid-frequency dynamics.

SUPPLEMENTARY NOTE 9: IMPACT OF CONTROL SOFTWARE CHANGES FOR UPS AT BUILDING 3

The UPS at Building 3 offers a unique opportunity for detailed analysis and monitoring. A wide range of measurements is available, including DC current measurements downstream of the three inverters and directly upstream of the batteries, as well as AC measurements both before and after the UPS, enabling isolation and characterization of the UPS behavior on the AC level itself. In addition, a power quality (PQ) meter (ID 023) is installed at the transformer level (see Table 1 and Fig. 5**a** in the main manuscript for an overview of the electrical layout). Furthermore, several operational parameters of the battery, in particular, the EMS, can be actively controlled, and we maintain continuous exchange with the manufacturers, allowing for deeper system insights and coordinated investigations.

The comprehensive measurement and control infrastructure enables targeted investigations of the UPS operating behavior and, in particular, the role of clock synchronization within the battery energy management system (EMS). To assess the influence of network time synchronization, we conducted comparative measurements under two conditions: (i) with NTP synchronization disabled and (ii) with NTP synchronization enabled on the battery EMS.

We find a remarkable difference between the operation with and without NTP synchronization. In both cases, the daily waveform measured at the Transformer progresses from day to day, but the magnitude of the progression differs considerably (Fig. 18a,b). Remarkably, the phase progression is much larger with NTP synchronization enabled, with a magnitude of approximately 5-7 seconds per day. The hourly waveforms of the power time series at transformer level provide further insights (Fig. 18c,d). We find that the pattern does not drift continuously, but jumps from day to day. A plausible explanation is that the EMS reschedules its operation every day. In summary, self-tests are done periodically in 1 min intervals, but the starting point is reset from day to day.

Based on our observations, the manufacturer of the UPS at Building 3 implemented a revised version of the algorithm used to estimate the battery's state of charge, as part of a maintenance intervention involving a software update for battery block 1. Following this update, the self-tests are no longer performed strictly every minute for battery block 1, see Fig. 31**a**. Instead, the tests occur at seemingly irregular time intervals, while the current is drawn over longer durations such that the overall consumed energy remains approximately constant. However, the start of the tests remains locked to the same second within each minute, resulting in a quasi-periodic behavior. The other two battery blocks are not affected, as they have not received an update yet.

This modification is clearly reflected in the Fourier spectrum of the DC current measured at battery block 1. Instead of a sharp spectral peak at a period of 1 min, the updated behavior produces a significantly broader spectral feature, see Fig. 31**b**. Nevertheless, such a broadband excitation may still interact with resonant processes in the power system and

thereby contribute to the emergence of a 1 min pattern in the grid frequency. The software update, therefore, represents a partial mitigation step toward eliminating the source of the observed signal, although it does not fully suppress the underlying synchronization effect.

SUPPLEMENTARY NOTE 10: THE 1 min PATTERN IN IRELAND

Ireland is special because of its small size with an annual consumption of 30,581 GWh in 2023 and a high penetration of data centres. The standard frequency range of the Irish system is ±200 mHz; in addition, the national regulator has put in place an incentive to keep the frequency within ±100 mHz for at least 98 % of the time [45]. The development of the electricity consumption and and the share of electricity consumed by data centres between 2016 and 2023 is summarized in the following Table based on raw data from [46].

Year	Metered Electricity Consumption per Year [GWh]			Share Data
	Data Centres	Other Metered Customers	Total	Centres [%]
2016	1,480	23,876	25,356	6
2017	1,760	23,966	25,725	7
2018	2,180	24,550	26,730	8
2019	2,488	24,017	26,505	9
2020	3,028	24,028	27,056	11
2021	4,010	24,496	28,506	14
2022	5,270	24,554	29,824	18
2023	6,334	24,246	30,581	21

We obtained frequency data from Ireland for 2016 with 1 s resolution directly from the network operator EirGrid, which is not publicly available and subject to a data-sharing agreement. Frequency data for 2021 was measured in the distribution grid in Dublin using an EDR. Low-resolution frequency data from 2026 has been scraped directly from EirGrid's Smart Grid Dashboard [8]. These data have substantially lower temporal and frequency resolution, with a sampling interval of 5 s and a frequency resolution of 10 mHz. A precise determination of the signal amplitude and its temporal evolution is therefore not possible. Instead, we compare representative data from three months in 2016, 2021, and 2026 (Supplementary Figs. 19-21).

For each month, we show the average minutely frequency profile over the complete month together with a selected one-hour interval in which the minute-scale signal is particularly pronounced and an extract of the corresponding raw frequency measurements in this hour. The monthly profiles indicate a substantial increase of the minute-scale component between 2016 and 2021, whereas the difference between 2021 and 2026 is considerably smaller. Given the limited frequency resolution of the 2026 measurements, the latter result should be interpreted qualitatively.

An especially pronounced example occurs on July 17, 2026, at 9 am (UTC). During this interval, the minutely profile exhibits a recurring frequency decrease of approximately 20 mHz at the beginning of the minute. The individual dips are also directly visible in the raw frequency measurements, demonstrating that the structure in the minutely profile is not

merely an artifact of averaging. During the same hour, the background frequency decreases over a period of approximately 20 min. The minute-scale perturbations are superimposed on this slower excursion and bring the measured frequency to approximately -100 mHz on four occasions, i.e., to the lower edge of the ± 100 mHz band, while remaining within the ± 200 mHz standard range. Because the measurements are quantized in steps of 10 mHz, it cannot be determined from these data whether the actual frequency remained just above or temporarily crossed this boundary.

SUPPLEMENTARY NOTE 11: IMPACT OF IOT AND SMART DEVICES

The adoption of household Internet-of-Things (IoT) devices is increasing worldwide [47]. Many of these devices rely on internal clocks synchronized via Internet-based timing protocols. As a consequence, large populations of distributed devices may execute internal control routines, status updates, or scheduled operations in a synchronized manner. Such synchronization can lead to quasi-periodic patterns in electricity consumption that may collectively contribute to the emergence of the 1 min pattern.

Empirical evidence for such synchronized behaviour is observed in commercially available smart household devices. For example, a smart baking oven whose internal clock is synchronized via the Internet exhibits a distinct 1 min pattern in its power consumption whenever the oven is operated using a timer function (cf. Fig. 22). The heating process is triggered at the beginning of each minute, with the heating demand determining the intensity of the power draw. This results in a stable, recurring 1 min power pattern. A Thermomix also displays recurring power peaks during operation; however, these do not follow a strict one-minute periodicity.

To further investigate synchronized timing effects in network-connected electronics, we analyzed the power consumption of four racks containing 12 Raspberry Pis each. To prevent the findings from being influenced by specific tasks performed with the devices, all Raspberry Pis remained unused during the tests. This allowed us to monitor their behavior and the corresponding power consumption due only to the basic operations of the operating system. All racks exhibit clear periodic signatures in their Fourier spectra, most prominently at 30 s, but also at 1 min and 2 min, see Fig. 23**a**. The corresponding time series reveal regularly recurring power spikes at the beginning of the minute, see Fig. 23**b**. These patterns remain stable over several days, indicating persistent and reproducible timing processes.

The analysis of the 30 s waveform further shows that three of the four racks are synchronized to the beginning of the interval, and thus also to the beginning of each minute, see Fig. 23**c**. Only one rack exhibits a persistent phase offset of approximately 20 s. These observations suggest that even relatively small networked computing devices can exhibit stable and synchronized timing behaviour over extended periods. While the individual power variations are small, the rapidly growing number of IoT and network-connected devices raises the possibility that their aggregated operation may contribute measurably to the 1 min frequency pattern.

These observations highlight the need for further investigation into the behaviour and control of smart devices, as their collective effects may pose additional challenges to frequency stability.

SUPPLEMENTARY NOTE 12: THE 1 min PATTERN IN MALLORCA

The strength of the 1 min pattern varies across power systems and fluctuates over time. Here, we analyse the development of the signal amplitude on the Spanish island of Mallorca from 2019 to 2023.

Mallorca is part of the Balearic synchronous area, which also includes the islands of Menorca, Ibiza, and Formentera. This grid is connected to the Continental European (CE) synchronous area only via a high-voltage direct current (HVDC) link [48], which allows for power transmission but does not establish frequency synchronization between the two systems.

In recent years, Mallorca's electricity supply has changed considerably. Two out of four generating units at the Es Murterar coal-fired power plant were withdrawn from regular operation at the end of 2019, and the operating hours of the remaining units were reduced [49]. In 2020, demand was covered primarily by combined-cycle gas turbines and imports via the HVDC connection [50]. Remarkably, the partial shutdown of the coal-fired power plant in Mallorca was accompanied by an overall *reduction* in frequency fluctuations [50].

Figure 24 shows weekly averages of the 1 min signal amplitude together with electricity generation from coal-fired power plants. The signal amplitude varies strongly and exhibits a pronounced negative correlation with coal generation. In fact, average amplitudes above 3 mHz are observed only during periods without coal generation. In contrast, amplitudes regularly drop below 2.5 mHz when coal-fired generating units are online. This observation is particularly remarkable given the simultaneous overall reduction in frequency deviations.

We interpret these findings as follows. On the one hand, the retirement of coal-fired generating units reduces rotational inertia, making the grid more susceptible to frequency fluctuations caused by rapid changes in the power balance. On the other hand, the HVDC converter station effectively contributes to secondary frequency control [51, 52], thereby reducing sustained frequency deviations. In addition, the converter aggressively suppresses frequency deviations exceeding 150 mHz [53]. As a consequence, slow and sustained frequency deviations are reduced, whereas faster fluctuations, including the 1 min pattern, become more pronounced. Together, these observations illustrate the impact of system inertia on the strength of the 1 min pattern.

SUPPLEMENTARY NOTE 13: IMPACT OF RISING AMPLITUDES ON FREQUENCY DYNAMICS IN THE SCANDINAVIAN GRID

In this supplementary note, we introduce a data-driven linear model of grid frequency dynamics and use it to quantify how an increase of the 1 min pattern amplitude affects frequency quality. The model parameters are inferred from frequency measurements by means of simulation-based inference, anchoring the study on realistic operating conditions. We use data from Scandinavia in 2025, cf. Fig. 33.

A. Methodology

1. *Grid model*

We describe the dynamics of the bulk frequency by the aggregated swing equation [14],

$$\begin{aligned} \frac{d\phi}{dt} &= \omega, \\ M\frac{d\omega}{dt} &= -c_1\omega - c_2\phi + \Delta P(t), \end{aligned}$$

where ϕ denotes the aggregated voltage phase angle, $\omega = 2\pi\,(f - f_{\mathrm{ref}}) = 2\pi\Delta f$ the angular velocity, with f the measured frequency and f_{ref} the reference frequency, which equals 50 Hz in the Scandinavian synchronous area, and M denotes the aggregated inertia. The aggregated swing equation describes the frequency response to a power imbalance ΔP, subject to damping and load-frequency control. In particular, the aggregated effects of damping, primary and secondary control are modelled by a PI law with droop parameters c_1 and c_2, respectively. Since the inertia M enters the dynamics only as a common scaling of the remaining terms, it cannot be identified separately from them. We therefore work with inertia-normalised quantities throughout and report effective parameters, e.g. $c_1 \equiv {}^{c_1}\!/_{M}$.

In Refs. [54, 55], the response to deterministic power imbalances induced by the market design and to stochastic fluctuations was analysed, described by a power step P_0, a linear power drift P_1, and Gaussian white noise ξ of amplitude ϵ. Here we extend the model by a periodic term P_{pattern}, which represents a disturbance recurring once every minute. Then, the power imbalance reads

$$\Delta P(t) = (P_0 + P_1 t) + P_{\mathrm{pattern}}(t) + \xi(t).$$

The empirical waveforms in Fig. 2 guide the choice of P_{pattern}. In every synchronous area the pattern consists of a cosine-shaped frequency deviation that recurs once per minute. What differs between the grids is how much of the minute the deviation occupies, and where within the minute it is located. In Ireland, for example, the deviation is completed

within roughly the first 10 s, so that the remaining 50 s are free of a systematic pattern, whereas in Continental Europe it extends over almost the entire minute and thus appears as a smoothed-out response with no pronounced quiet interval. To capture this range of behaviours with a single functional form, we define

$$P_{\text{pattern}}(t) = \begin{cases} A\cos\left(2\pi\left(\dfrac{\tau(t)}{T_{\cos}} + \dfrac{\Delta t_{\text{off}}}{T}\right)\right), & \tau(t) < T_{\cos}, \\ 0, & \tau(t) \geq T_{\cos}, \end{cases}$$

where

$$\tau(t) = (t - t_{\text{s}}) \bmod T$$

is the phase within the minute, with $T = 60\,\text{s}$.

The four parameters control complementary aspects of the waveform. The amplitude A scales the magnitude of the deviation. The cosine period $T_{\cos}$ sets how much of the minute is occupied by the deviation and thereby interpolates between two regimes. For $T_{\cos} < T$, a complete cosine cycle of length $T_{\cos}$ is followed by a quiet interval of length $T - T_{\cos}$, which reproduces the sharp Irish response. For $T_{\cos} \geq T$, the quiet interval vanishes and the cosine is truncated at the minute boundary, so that the pattern is active throughout the minute and only the fraction $T/T_{\cos}$ of a full cycle is traversed before the pattern restarts. The onset time t_{s} positions the deviation within the minute, capturing patterns that are not aligned with the start of the minute, and the phase shift $\Delta t_{\text{off}}/T$ selects which section of the cosine cycle is traversed, which determines whether the deviation appears as a drop, a rise, or a combination of both. Together, these degrees of freedom cover the observed variety of waveforms, all of which produce a peak in the Fourier amplitude spectrum at a period of one minute.

2. NPE model for parameter inference

To obtain realistic parameters for the grid model, we employ neural posterior estimation (NPE), a simulation-based inference technique [56–58]. Following the approach of our previous work [55], the workflow is sketched in Fig. 25.

During training, parameter sets are drawn uniform priors and the corresponding frequency trajectories are generated by integrating the aggregated swing equation with an Euler–Maruyama scheme. A conditional normalizing flow is then trained on these pairs of simulated trajectories and ground-truth parameters, learning the mapping to the normal distribution. All parameters and their prior distributions are summarized in the following Table.

Start value	Parameter	Prior distribution
$\phi(0)$ [rad]		$\mathcal{U}(-1.25, 1.25)$
$\omega(0)$ [rad/s]		$\mathcal{U}(-\pi, \pi)$
	c_1 [s^{-1}]	$\mathcal{U}(0.001, 0.08)$
	c_2 [s^{-2}]	$\mathcal{U}(0.00001, 0.0003)$
	P_0 [s^{-2}]	$\mathcal{U}(-0.05, 0.05)$
	P_1 [s^{-3}]	$\mathcal{U}(-0.00005, 0.00005)$
	ϵ [$\mathrm{s}^{-3/2}$]	$\mathcal{U}(0, 0.05)$
	A [s^{-2}]	$\mathcal{U}(0, 0.05)$
	$T_{\cos}$ [s]	$\mathcal{U}(35, 63)$
	t_{s} [s]	$\mathcal{U}(-30, 30)$
	Δt_{off} [s]	$\mathcal{U}(-30, 30)$

Once trained, the inverse of the normalizing flow can be used to infer realistic parameter distributions for measured data: conditioning it on an empirical observation, here a 15 min segment of the recorded frequency time series, and sampling from a standard normal base distribution yields posterior distributions over the model parameters. In particular, we apply the trained estimator to 27 301 non-overlapping segments I_i of length 15 min each, taken from the Scandinavian frequency recording between 19 March and 31 December 2025 at a sampling rate of 1 Hz. For every segment, we draw 100 posterior samples and use the posterior mean as the point estimate.

3. Counterfactual variation of the pattern amplitude

To quantify how a rising pattern amplitude affects the frequency dynamics, we use the grid model to simulate new frequency trajectories for each of the 27 301 segments I_i, keeping all inferred parameters fixed at their posterior means and varying only the amplitude A of the one-minute disturbance. Specifically, we scale the inferred amplitude of each segment by factors of 1 to 10, so that every scenario preserves the realistic intraday variation of the pattern and differs only in its magnitude, with the factor 1 serving as the reference that reproduces the measured conditions. Since the remaining dynamics stay at their inferred values, the resulting trajectories constitute a counterfactual study that isolates the contribution of the pattern to the frequency dynamics.

To assess the effect of increasing A on frequency quality, we evaluate two indicators for every segment and every scenario. The first is the largest frequency deviation in either direction, hereafter referred to as the Nadir for simplicity,

$$\mathrm{Nadir}(I_i) = \Delta f\left(\arg\max_{t \in I_i} |\Delta f(t)|\right),$$

and the second is the mean squared deviation

$$\mathrm{MSD}(I_i) = \frac{1}{|I_i|} \sum_{t \in I_i} \Delta f(t)^2,$$

with $|I_i| = 900$ samples per segment.

In addition to the two indicators, we compare the distributions of the frequency deviations and of the 10 s frequency increments across the scenarios. All quantities are computed for the empirical time series as well, which allows the simulated reference scenario to be checked against the measurements.

B. Results

1. NPE model parameters and validation

We validate the inference in two steps. First, we compare the daily profile of the inferred pattern amplitude A with the amplitude obtained directly from the measurements, see Fig. 26. The two intraday trends agree closely, confirming that the neural posterior estimator recovers the pattern amplitude reliably. Furthermore, the daily profiles of the remaining parameters also exhibit a pronounced time-of-day dependence, see Fig. 27, and are consistent with the estimates reported in Ref. [55]. Second, trajectories simulated with the inferred parameters reproduce both the visual character and the autocorrelation structure of the measured signal, see Fig. 28.

2. Impact of increasing amplitudes on frequency dynamics

The two stability indicators respond differently to a growing amplitude as shown in Fig. 29: the Nadir increases approximately linearly with A, whereas the MSD follows a quadratic trend, as expected for a quantity that is quadratic in the frequency deviation.

Beyond the aggregated indicators, an increasing amplitude also reshapes the full distributions. Figure 30a shows that a larger pattern amplitude broadens the distribution of frequency deviations and, in particular, substantially increases the number of large deviations relative to the empirical time series. Correspondingly, Fig. 30b shows markedly larger frequency increments on a 10 s timescale. For a fivefold increase of the amplitude with respect to the value inferred from the measurements, the standard deviation of the 10 s increments grows by a factor of 1.5.

Taken together, the growth of the Nadir, and in particular of its upper tail, and the broadening of the frequency statistics point to the additive nature of frequency deviations: the deterministic pattern enters the power imbalance alongside the stochastic fluctuations and the market-driven step and drift, so all contributions superpose. At the amplitudes (up

to $10A$) studied here, the 60 s pattern alone does not constitute an imminent threat to system stability, as the deviations remain within normal operating conditions. Its relevance lies in the superposition: a large pattern amplitude coinciding with a large stochastic fluctuation or a market-boundary ramp yields larger frequency deviations, and correspondingly larger Nadirs, than either contribution alone. The effect is therefore most pronounced in the tail of the Nadir distribution, where the deviations approach the limits of normal operation.

Overall, the model captures the joint effect of frequency control, power imbalance, short-term fluctuations, and the one-minute disturbance amplitude on the frequency deviations, and thereby provides a starting point for more elaborate nonlinear descriptions [59]. A natural next step is to relate the inferred parameters, and the pattern amplitude in particular, to techno-economic features of the power system.

SUPPLEMENTARY NOTE 14: EDR DEVICES

The Electric Data Recorder (EDR) described in Ref. [6] samples the voltage waveform with 25 kHz. We align the samples to UTC via a GPS reference signal, and filter them with a 200-sample Hamming window. From this oscillogram-like trace we interpolate the timestamps of the zero-crossings (ZC). Zero-crossing with inclining and declining voltages are separated. From the difference of subsequent ZCs with positive slope we determine the duration of the periods to arrive at the frequency estimates. Averaging these estimates over the desired reporting interval (1 second in this case) yields frequency time-series. Since the grid frequency changes continuously, the ZC timestamps do not add up to full seconds (and are shifted against UTC). Periods that overlap the left or right edge of the reporting window are therefore included in the average with a weighing factor. This factor is determined according to the overlap, assuming a constant frequency during the period. See Ref. [60] for further details and experimental verification of the method under dynamic conditions.

In order to verify the EDR frequency time series to other measurement methods, we compare EDR recordings to openly available data from German TSOs as well as to measurements by the PQ meters. We perform a comparison of these 3 different measurements in the Continental European synchronous area for a period of 15 days, ranging from June 28, 2025 to July 12, 2025. The EDR measurement is recorded in Karlsruhe while the frequency measurements by PQ meter are recorded in Jülich. The EDR as well as the measurements by the German TSOs have a sample rate of one second, while the PQ measurements have a sample rate of three seconds. In order to compare the data, we sample all three time series down to a sample size of three seconds.

As observed in Ref. [34] the frequencies differ only minimally within the Continental European Synchronous Area. We note that the various frequency measurements yield very similar frequency data series. In Fig. 32 we observe that the Fourier spectrum looks approximately identical for the three measurement recordings that are obtained by different signal processing techniques. Furthermore, we evaluate the Pearson correlation coefficients for the different frequency time series recorded with a PQ Meter in Jülich, recorded with an EDR in Karlsruhe and provided by a German TSO and find the following results.

	PQ Meter Jülich	German TSOs	EDR Karlsruhe
PQ Meter Jülich	1.000	0.979	0.991
German TSOs	0.979	1.000	0.969
EDR Karlsruhe	0.991	0.969	1.000

It is evident that the frequency recordings are strongly correlated. These observations show that the EDR measurements provide reliable time series that are particularly useful for our analysis.

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